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A comprehensive survey on communication optimization and privacy preservation in federated learning

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Abstract
A collaborative paradigm, federated learning enables devices to train machine learning models locally, without exposing raw data. The server initiates the training by transmitting the global model parameters to the clients, where it is trained using the local data. The model updates are communicated to the server, which in turn aggregates the model parameters and generates the global model. Multiple federated learning cycles are enacted until the model converges. Although data retention occurs at the client, the frequent exchange of local model parameters with the server poses notable privacy concerns and communication overhead. To address these challenges, an array of state-of-the-art mechanisms have been developed to safeguard user privacy and improve communication efficiency. In particular, privacy preserving measures can induce communication latency. Despite considerable advancements in this field, inherent trade-offs between accuracy, communication overhead, and privacy protections persist in hindering the development of FL frameworks. In this survey, we optimize our focus on approaches to improve communication efficiency while preserving the privacy of user data. This underscores the importance of exploring hybrid methods that effectively mitigate the dual concerns of privacy and communication. Additionally, we presented a decision framework for choosing communication efficient and privacy preserving FL strategies based on system limitations and threat models, as well as a unified compatibility matrix of these techniques in FL.

Keywords Federated learning, Communication efficiency, Differential privacy, Homomorphic encryption, Secure multiparty computation

1 Introduction

Federated Learning (FL) is an innovative approach in machine learning that enables decentralized model training, where the data is kept on local devices rather than being stored on a central server. Originally introduced by Google in 2017 [1], FL was designed as a privacy-preserving solution to address the risks associated with traditional centralized machine learning, which typically involves collecting data on a central server for training. In contrast to this method, FL enables various devices or organizations,

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including smartphones, hospitals, or financial institutions, to collaboratively train a common model without transferring sensitive data. This methodology is beneficial in sectors with rigorous privacy mandates, such as healthcare and finance, where data must be retained at its origin.

The motivation behind FL is centered at the growing need to protect user privacy, enhance data security, comply with data privacy regulations, and facilitate collaborative learning across diverse and inaccessible data sources. By allowing data to stay on the local device, FL mitigates the risks of data exposure while enabling machine learning on a broad scale. Although FL reduces communication overhead and privacy concerns in transmitting entire data to the server, frequent exchange of model updates and global model parameters induces communication overhead and data leakage. Another key challenge in FL is the heterogeneity of the system and data.

The primary goal of this survey is to provide a comprehensive introduction to FL, including its fundamental ideas, approaches, and important applications. This survey delves into fundamental FL concepts, privacy and security techniques, and popular FL algorithms and models. Through the discussion of privacy preserving methods, including Homomorphic Encryption (HE) [2], Secure Multi-Party Computation (SMPC) [3], and Differential Privacy (DP) [4], this overview gives the reader a detailed understanding of how FL protects user data. In addition to privacy concerns, communication overhead is also a challenge in FL. After each FL round, the client and server exchange the model parameters, resulting in a higher bandwidth requirement, specifically for a heavy deep learning model with more complex layers. In addition, it explores the applications of FL in various domains, such as healthcare [5–7], finance, image analysis [8], IoT, and autonomous vehicles, highlighting the specific advantages that FL brings to each field. The survey also investigates FL trade-offs, such as the balance of model accuracy, privacy preservation, communication efficiency, and computational costs.

By emphasizing the current developments and open research challenges in FL, such as handling Non-Independently and Identically Distributed (Non-IID) data [9], improving scalability, and enabling personalization, this survey aims to equip researchers and practitioners with a comprehensive understanding of the potential of FL. This overview serves as a foundation for further research and innovation in FL, ultimately fostering its development and impact across various sectors. The primary contributions are listed below.

- A detailed categorization of FL is explained on the basis of various aspects.
- A detailed analysis of potential threats to FL and its defensive mechanisms of a secure FL framework is covered.
- The causes of communication overhead in FL and its cutting-edge remedies have been thoroughly examined.
- Methods to preserve privacy in FL will result in communication overhead. Therefore, we thoroughly examine the most recent approaches to create an FL framework that safeguards privacy while facilitating efficient communication.
- The requirements for communication and privacy in different FL applications are examined.

The rest of the paper is as follows. In Sect. 2, we present the survey protocol. In Sect. 3, we describe the overview of FL. Privacy in FL is presented in Sect. 4, followed by

Communication Efficiency in FL in Sect. 5. In Sect. 6, we explore the optimization of privacy and communication efficiency. Section 7 details the impact of heterogeneity on communication and privacy. Section 8 explains the contextual analysis of communication threat defense tradeoffs in FL. Section 9 explores the practical challenges in establishing communication efficient and secure FL systems. Finally, we conclude the paper and outline future directions in Sect. 10. The sections-wise organization of the manuscript is illustrated in Fig. 1.

1.1 Distinct contributions compared to existing surveys

This survey presents a comprehensive perspective by jointly considering privacy preservation and communication efficiency, both of which critically influence the feasibility and scalability of FL systems. This study provides a detailed and structured analysis of the privacy aspect of FL, which systematically combines many dimensions. Unlike other surveys that primarily focus on the taxonomy of attacks or isolated defense mechanisms, this survey presents a unified analysis of privacy attacks together with their corresponding defense strategies. In particular, our work provides a structured analysis of various types of privacy attacks and further focuses on providing a detailed overview of their corresponding defenses. Moreover, it expands the scope of analysis beyond the commonly studied dimensions by discussing some of the widely used privacy preserving

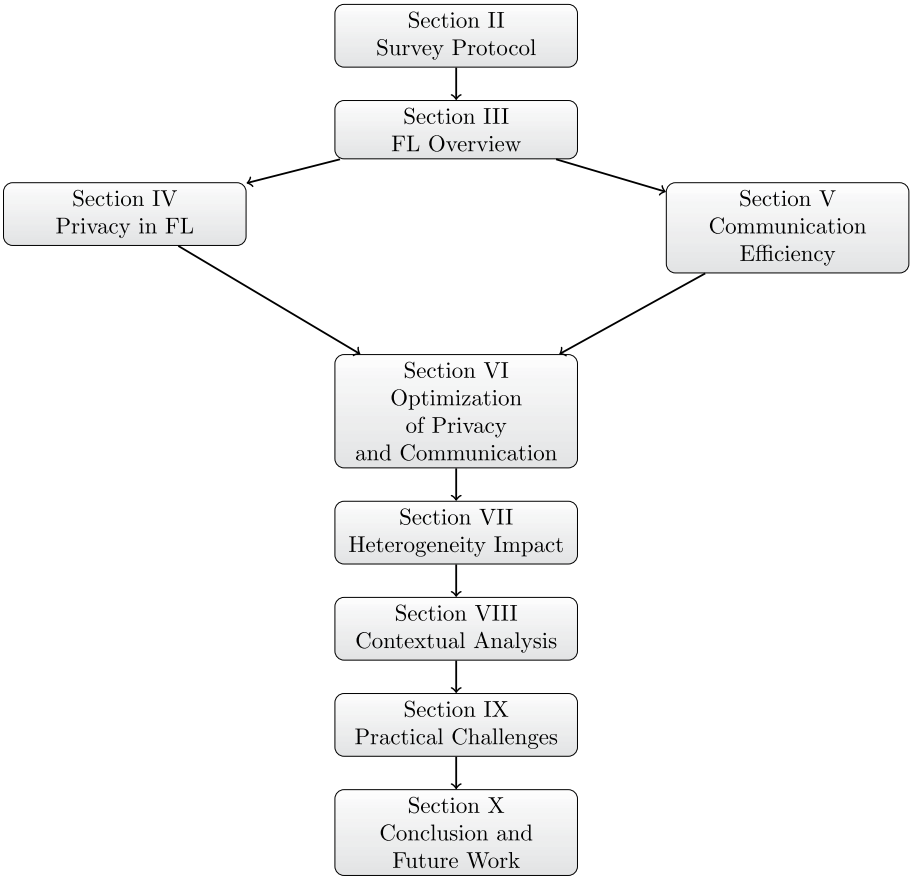


Fig. 1 Schematic overview of the survey structure. The framework progresses from foundational concepts (FL overview) to core pillars (privacy and communication efficiency), followed by their joint optimization and system-level factors such as heterogeneity and application-driven contextual analysis and finally practical deployment challenges and future directions

techniques and analyzing their effects on the performance of the system. In addition, it discusses the trade-off between privacy preservation and model utility with respect to the accuracy, efficiency, and robustness of a system.

Existing surveys frequently regard communication as a peripheral issue; nevertheless, our research prioritizes it and analyzes it from both algorithmic and system-level viewpoints. Specifically, communication-efficient FL techniques are organized into a structured taxonomy that includes model compression, local computation optimization, communication protocol design, client selection, hybrid FL approaches, and additional strategies. This unified categorization clarifies the design space and emphasizes the mutual dependence among various methods, which have not been fully discussed in previous surveys.

Moreover, this survey provides an integrated analysis framework that jointly considers communication cost, privacy assurances, and model utility. It also introduces a compatibility matrix and decision framework to facilitate the selection of appropriate FL mechanisms under varying deployment constraints. In addition, this survey discusses practical deployment challenges and emerging paradigms such as event-driven communication, over-the-air computation, and hybrid FL systems. These contributions collectively distinguish this work as a comprehensive reference for both researchers and practitioners working on communication efficient and privacy preserving FL systems.

Table 1 compares this survey with recent high-impact FL surveys from reputed journals. The comparison highlights major differences in terms of taxonomy scope, treatment of communication efficiency and privacy jointly, attack-defense analysis, heterogeneity considerations, system-level trade-off analysis, deployment-oriented insights, and support for practical decision-making.

2 Survey protocol

This survey adheres to a controlled literature selection approach based on PRISMA principles to guarantee transparency and reproducibility. We thoroughly searched many academic databases, such as Scopus and Google Scholar, to find studies that were relevant. The literature search utilized combinations of terms related to FL, security, privacy, and communication efficiency. An example of a search string is ("federated learning" OR "FL") AND ("security" OR "privacy" OR "attack" OR "defense" OR "compression" OR "communication efficiency"). The search was limited to publications from 2017 to 2026 so that it could find the most recent developments in the subject. The first search found a lot of publications, and the first step was to look through them based on their titles and abstracts. After removing duplicate entries, the remaining studies were fully reviewed to see if they met the inclusion criteria. This meticulous screening left us with a final corpus of 187 publications that we employed for analysis.

A systematic review strategy was used to search for articles of relevance in different sources. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) diagram that shows the identification, screening, and inclusion of studies in this survey is depicted in Fig. 2. This search found 17,303 articles in Scopus, and 56 more articles were found in other databases. After reviewing titles and abstracts, 1847 full-text articles were reviewed, leading to the final selection of 187 articles for this study.

The total number of articles reviewed annually is illustrated in Fig. 3. The trend points towards an unprecedented rise in the number of research investigations conducted

Table 1 Comparison of this survey with representative federated learning surveys from reputed journals

Survey	Privacy analysis	Communication efficiency	Attack & defense analysis	Trade-off analysis	Heterogeneity analysis	System-level deployment insights	Hybrid/adaptive FL	Decision framework/compatibility
[10]	Limited	✓	×	Limited	✓	✓	Limited	×
[11]	Limited	✓	×	Moderate	Moderate	✓	Limited	×
[12]	Limited	✓	×	Limited	Limited	Moderate	×	×
[13]	✓	Limited	✓	Limited	Limited	Moderate	×	×
[14]	✓	×	✓	Limited	Limited	Moderate	×	×
[15]	✓	×	✓	Limited	Moderate	Limited	×	×
[16]	✓	Limited	Moderate	Moderate	Moderate	Moderate	Limited	×
[17]	Moderate	✓	Limited	Moderate	✓	Moderate	✓	×
[18]	✓	✓	Moderate	✓	✓	✓	✓	Limited
Ours	✓	✓	✓	✓	✓	✓	✓	✓

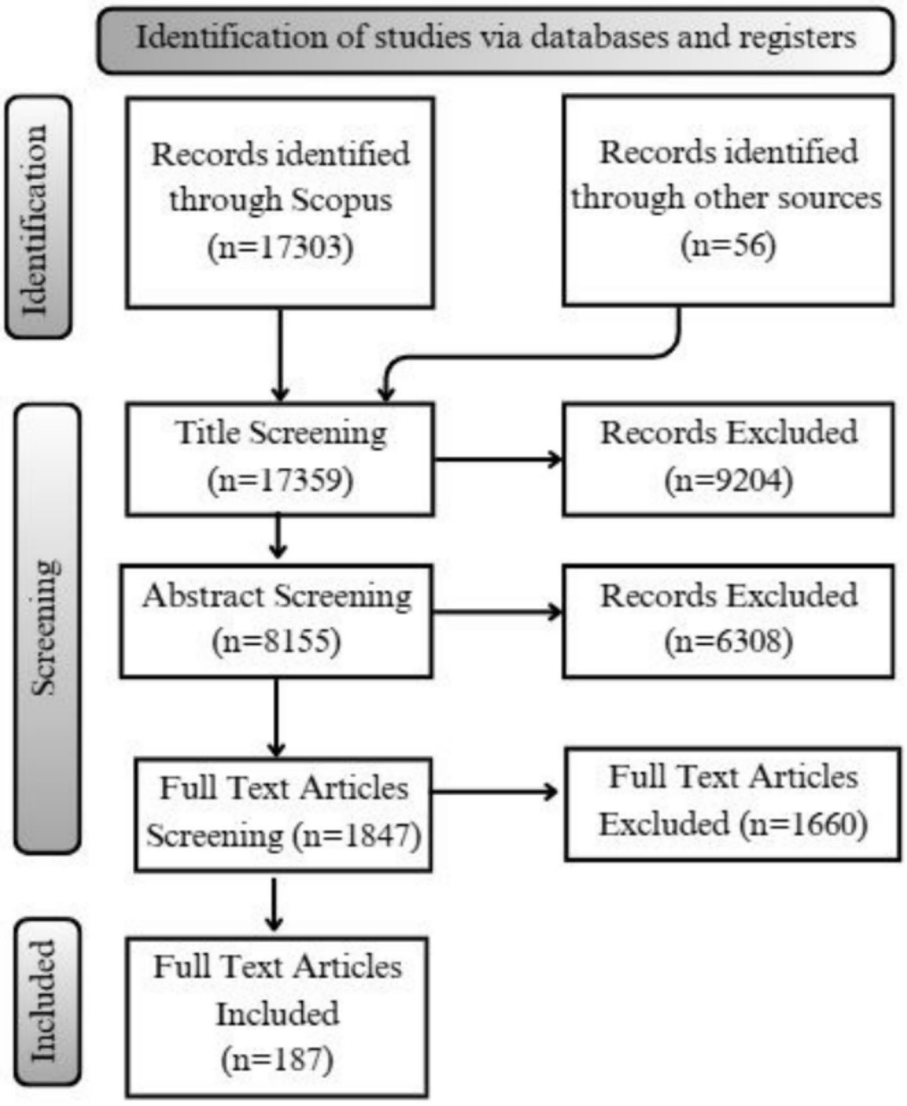


Fig. 2 Prisma diagram

within this field of study in the last few years, with both 2024 (39 articles) and 2023 (43 articles) being the high points. The distribution points towards a growing curiosity in the area, especially after 2021, and an indication of the subject's importance in the research community.

The articles included were divided into four broad categories: privacy, communication, FL, and both communication and Privacy. The results of categorization are presented in Fig. 4, with most studies focusing on Privacy (106 articles), followed by communication (52 articles), FL (22 articles), and communication and privacy (7 articles). These results indicate that privacy issues and communication efficiency are among the most dominant in research discourse in this area.

3 Overview of federated learning

An FL System follows client–server topology, where the server acts as the central component, which distributes the global model parameters to all participating clients, which then perform the local model training for 'e' epochs and share the local updates back to

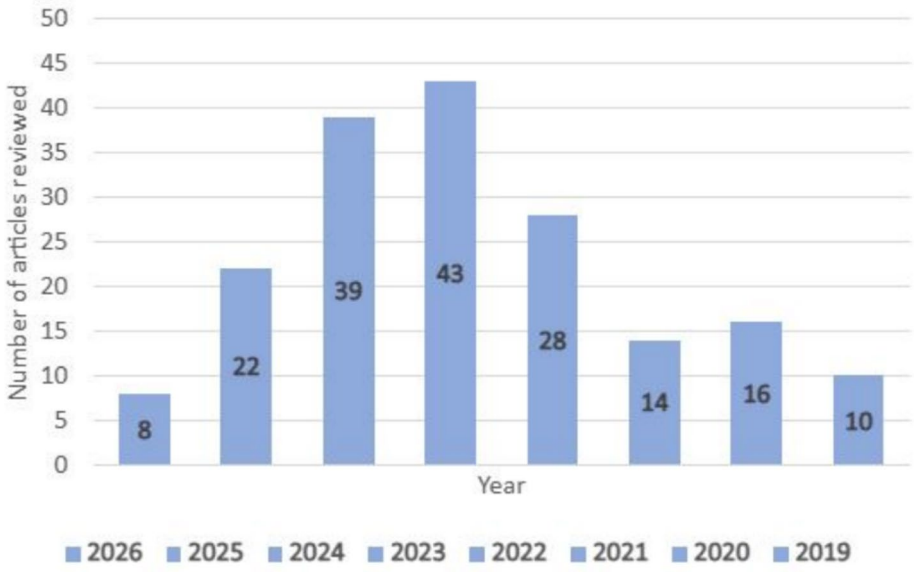


Fig. 3 Number of articles reviewed per year

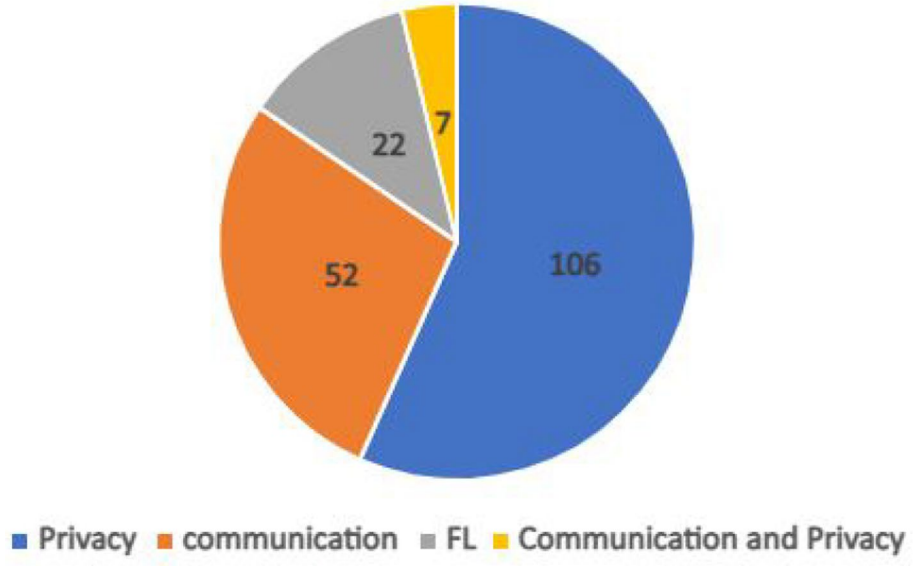


Fig. 4 Number of articles reviewed on each category

the server. The server then aggregates the model updates to generate the global model for the next FL round. This process is repeated several times until the model converges. Client devices, which store data and conduct training locally, are involved in the process. This ensures that the data stays on the client device. The FL framework is shown in Fig. 5, and the terminologies and abbreviations used in the paper are shown in Tables 2 and 3 respectively.

FL seeks to derive a global model by optimizing a distributed objective function across several clients.

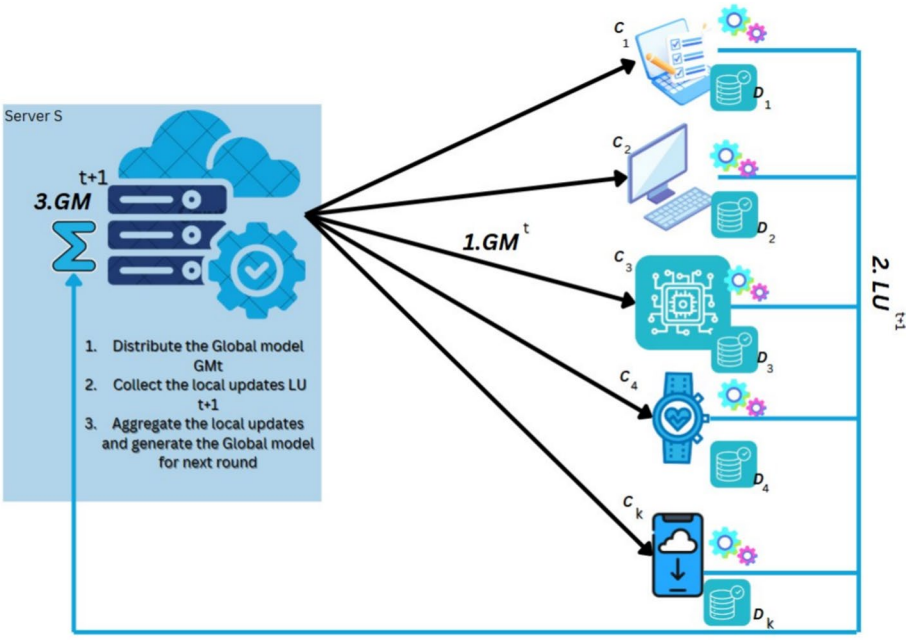


Fig. 5 Federated learning overview

$$F(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \tag{1}$$

where $F_k(w)$ is the local objective function of client k, n_k is the number of data samples at client k, and n= $\sum_{k=1}^{K} n_k$. The Federated Averaging (FedAvg) algorithm is utilized to minimize the global objective function *F*(*w*). The global model is updated as follows:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^t \tag{2}$$

where w_k^t represents the model that client k updated locally at round t. This concept is the basis for several types of FL structures.

3.1 A taxonomy of federated learning types

Figure 6 displays the taxonomy of FL types according to several criteria. In FL, the client devices independently capture the data and train the global model using their local dataset. The distribution of data across client devices plays a critical role in FL. Based on data distribution, FL can be classified into three types: Horizontal FL, Vertical FL, and Federated Transfer Learning(FTL). In horizontal FL [19], all client datasets share the same set of features but contain different samples. This is similar to slicing a dataset horizontally and distributing it to several clients. The local loss function at each client is

$$F_k(w) = \mathbb{E}_{(x,y)\sim D_k}[\ell(w; x, y)] \tag{3}$$

where D_k denotes the local dataset of client *k*. In contrast, vertical FL [20] client datasets have a different set of features for the same set of samples. The learning objective depends on the combination of features distributed across multiple clients and can be expressed as:

Table 2 Terminologies

Notation	Meaning
S	Server
c_k	k^{th} Client
D_k	Local dataset of Client c_k
e	Number of local Epochs in each FL round
GM^t	Global model at t^{th} FL round
LU_k^{t+1}	Local update send by Client c_k after training GM^t using D_k
d_k^p	p^th share of data held by client c_k
y	Label space
F(w)	Distributed objective function to be optimized by the global model
$F_k(w)$	Local objective function to be optimized by client k
n_k	Number of data samples at client k
N	Total Number of data samples used by all the participating clients
w^t	Global model parameters at round t
w_k^t	Local model that client k updated locally at round t
x_k	Feature subset held by client k in Vertical FL
$\ell(w; x, y)$	Local Loss Function at client k
$L(w; x, y)$	Global loss Function
$\mathcal{N}_k$	set of neighboring clients of client k in distributed FL
a_{kj}	the aggregation weights at client k corresponding to each neighbour j in distributed FL
$\mathcal{G}_g$	The set of clients in group *g* in hierarchical FL
w_g^t	Intermediate aggregated models at round t in hierarchical FL
α_t	The mixing factor in asynchronous FL to mitigate the staleness
D, D'	Neighboring datasets differing in one data point
$\mathcal{M}$	Randomized mechanism
$\mathcal{S}$	Output space of the mechanism
$\Pr[\cdot]$	Probability of an event
ϵ	Privacy loss parameter
δ	Failure probability
w_k	Local model update of client *k*
$\tilde{w}_k$	Noisy local update
ξ_k	Client-side noise, $\xi_k \sim \mathcal{N}(0, \sigma^2)$
σ^2	Variance of Gaussian noise
K	Number of clients
D_G	Aggregated model update, $D_G = \frac{1}{K}\sum_{k=1}^{K} \tilde{w}_k$
$\tilde{D}_G$	Noisy aggregated model
ξ_s	Server-side noise, $\xi_s \sim \mathcal{N}(0, \sigma^2)$
$f(\cdot)$	Joint computation function
$\mathbb{D}_i$	Input domain
$\mathbb{Y}$	Output space
$h(\cdot)$	Function on aggregated shares
y	Final output
$y^{(i)}$	Partial output share

$$F(w) = \mathbb{E}_{(x_1, x_2, \ldots, x_K, y)\sim D}[L(w; x_1, x_2, \ldots, x_K, y)] \tag{4}$$

where x_k represents the feature subset held by client *k*, requiring collaborative computation of the loss function. In FTL [21], both the sample and the feature space differ across the clients. FTL enables collaboration by leveraging knowledge transfer, allowing a model trained on one client's dataset to enhance the training process on another client's dataset, particularly when data overlaps are minimal. Consequently, each client optimizes a distinct local objective:

Table 3 Abbreviations

Abbreviation	Description
FL	Federated Learning
HE	Homomorphic Encryption
SMPC	Secure Multi-Party Computation
DP	Differential Privacy
Non-IID	Non independently and identically distributed
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
FTL	Federated Transfer Learning
IID	Independent and Identically Distributed
RQSGD	Revised Quantized Stochastic Gradient Descent
P2P	peer-to-peer
SL	Split Learning
KD	Knowledge Distillation
MIA	Membership Inference Attack
DLG	Deep Leakage from Gradients
iDLG	Improved Deep Leakage from Gradients
WDLG	Wasserstein distance-based Deep Leakage from Gradients
HCGLA	Highly compressed gradient leakage attack
GAN	Generative Adversarial Network
GMI	Generative Model Inversion
GSI	Gradient Sign Inversion
CIP	Client Level Input Perturbation
CoAE	Confusional AutoEncoders
PEFL	Privacy-Encoded Federated Learning
KDRSFL	Knowledge Distillation Resistance Transfer for Split Federated Learning
SFL	Split Federated Learning
LID	Local Intrinsic Dimensionality
PASS	Parameter-Audit Based Secure and Fair FL Scheme
LDP	Local Differential Privacy
GDP	Global Differential Privacy
LCDP	Local Central Differential Privacy
CDP	Central Differential Privacy
PHE	Partial Homomorphic Encryption
FHE	Fully Homomorphic Encryption
SWHE	Somewhat Homomorphic Encryption
TEE	Trusted Excecution Enviornment

$$F_k(w_k) = \mathbb{E}_{(x,y)\sim D_k}[\ell_k(w_k; x, y)] \tag{5}$$

where model parameters and data distributions may vary across clients, and knowledge is transferred through shared representations or model components.

Based on the nature of the client device, FL systems can be grouped into Cross Silo and Cross Device. Cross silo FL involves a small number of reliable clients, such as organizations, where each client device has a large and reliable dataset [22]. Here the FedAvg update Eq. (2) is computed over a small set of K clients with relatively large n_k, leading to stable and consistent model updates. Cross device FL involves a large number of participating clients with smaller and less reliable datasets [23]. Due to the large K and limited client availability, only a subset $\mathcal{S}_t \subseteq \{1, \ldots, K\}$ typically participates in each training round. Therefore, the aggregation is approximated as:

$$w^{t+1} = \sum_{k\in\mathcal{S}_t} \frac{n_k}{\sum_{j\in\mathcal{S}_t} n_j} w_k^t \tag{6}$$

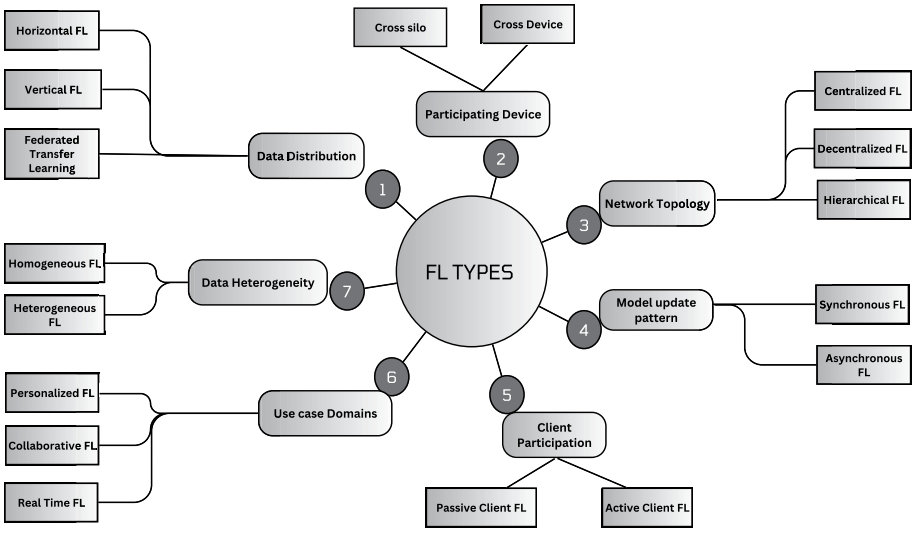


Fig. 6 Taxonomy of Federated Learning based on distribution of data, participating device, Network topology, Model update pattern, Client participation, Use case Domain and Data Heterogeneity

which introduces additional challenges such as partial participation, statistical heterogeneity, and variability in updates.

Based on the underlying network structure, FL can be divided into three main types: centralized, decentralized, and hierarchical FL. A central server coordinates the training process in centralized FL by collecting and combining model updates from all participating clients. The aggregation follows pure FedAvg algorithm as described in Eq. 2. The server is the sole repository for all changes, rendering it susceptible to a single point of failure. Decentralized FL, on the other hand, does not need a central server [24, 25]. Instead, clients communicate local model parameters to their neighbors directly and work together to aggregate models in a distributed way, as shown in Eq. 7,

$$w_k^{t+1} = \sum_{j \in \mathcal{N}_k} a_{kj}\, w_j^t \tag{7}$$

where $\mathcal{N}_k$ denotes the set of neighboring clients of client k, and a_{kj} represents the aggregation weights that satisfy $\sum_j a_{kj} = 1$. This removes the single point of failure but introduces communication and synchronization challenges. This basic difference means that there are different trade-offs when it comes to communication overhead, robustness, and scalability. In hierarchical FL, clients are split into several groups. Each group has an intermediate aggregator that collects local updates and sends them to a higher-level server or aggregator [26]. The aggregation can be expressed in two stages:

$$w_g^t = \sum_{k \in \mathcal{G}_g} \frac{n_k}{n_g} w_k^t, \quad w^{t+1} = \sum_{g=1}^{G} \frac{n_g}{n} w_g^t \tag{8}$$

where $\mathcal{G}_g$ represents the set of clients in group g, and w_g^t is the intermediate aggregated model. This hierarchical aggregation reduces communication overhead while maintaining a structured coordination mechanism.

Model aggregation can be done once in an FL round after receiving all local updates or several times in an FL round after receiving each local update. The former is called

synchronous FL, and the latter is asynchronous FL [27]. Hybrid or semi-asynchronous [28] approaches balance the benefits of both synchronous and asynchronous FL. In synchronous FL, the server waits for all selected clients to complete their local training before performing aggregation, typically following the standard FedAvg update as in Eq. 2. In asynchronous FL, the global model is updated immediately upon receiving each client update, without waiting for others:

$$w^{t+1} = (1 - \alpha_t)w^t + \alpha_t \, w_k^t \tag{9}$$

where w_k^t is the update from client k and α_t is a mixing factor that may depend on staleness or delay.

In semi-asynchronous or hybrid FL, aggregation is performed over a subset of available clients or within a bounded delay, combining aspects of both approaches:

$$w^{t+1} = (1 - \beta)w^t + \beta \sum_{k \in \mathcal{S}_t'} \frac{n_k}{\sum_{j \in \mathcal{S}_t'} n_j} \, w_k^t \tag{10}$$

where $\mathcal{S}_t' \subseteq \mathcal{S}_t$ represents the subset of clients whose updates are received within a predefined time window. The parameter $\beta \in (0, 1)$ is a mixing coefficient that controls the contribution of the aggregated updates from a subset of participating clients relative to the previous global model.

FL involves multiple participating devices. Based on the participation of client devices in training, FL can be classified as active client FL and passive client FL. In active client FL [29, 30], only a fraction of clients actively participate in the training phase. This subset can be identified using a variety of variables, such as device data volume, communication bandwidth, and computational resources. The global model is updated using only the selected clients:

$$w^{t+1} = \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j} \, w_k^t \tag{11}$$

In passive client FL, all clients contribute to the training process irrespective of their communication bandwidth and computational resources. The aggregation strategy is the same as FedAvg as described in Eq. 2.

FL is applied across various domains, primarily categorized into personalized, collaborative, and real-time applications, each addressing different needs while preserving data privacy. Personalized FL [31] is dedicated to personalizing models for individual users or devices and leveraging collaborative learning. It is extensively applied in health care for customized diagnostics and wearable health tracking, mobile applications such as Google's Gboard for adaptive text prediction, and online shopping for customized product recommendations. A Reinforcement Learning (RL) driven adaptation to handle client heterogeneity and dynamically personalize models is specified in [32]. Collaborative FL [33] allows different organizations to train models collaboratively without data exposure, which is suitable for healthcare and genomics, where hospitals cooperate in training predictive models to detect diseases, finance, where banks create fraud models without exposing customer information, and autonomous cars, where manufacturers improve self-driving models while data privacy is guaranteed. Models must undergo continual real-time modifications in real-time FL [34], as the data is not trained centrally

but evolves perpetually. Real-time FL is applicable in IoT and Edge AI. It is used in smart city anomaly detection, cyber security in detecting threats in real-time, which enhances defense processes; and autonomous systems that keep updating models of navigation utilizing sensor data in a local network. These applications illustrate FL's flexibility in distributed machine learning that preserves privacy and is thus vital across sectors.

In FL, data heterogeneity describes differences in data distribution among various clients, which has a significant effect on model training and convergence. With homogeneous or Independently and Identically Distributed (IID) data, all the clients have data samples from the same distribution.

$$D_k \sim \mathcal{D}, \quad \forall k \in \{1, \ldots, K\} \tag{12}$$

As a result, the local objective functions are similar, i.e., $F_k(w) \approx F(w)$, leading to consistent updates and faster convergence, allowing standard aggregation methods such as FedAvg as in Eq. 2, to perform effectively.

However, in most practical applications, FL is faced with heterogeneous or Non-IID [35] data, where clients do not share similar data distributions due to reasons such as skew of the feature distribution, label imbalance, different data sizes, and concept changes.

$$\mathcal{D}_i \neq \mathcal{D}_j \text{ for } i \neq j \tag{13}$$

This discrepancy leads to divergence in local gradients:

$$\nabla F_k(w) \neq \nabla F_j(w) \tag{14}$$

which can cause slower convergence, model drift, and degraded global performance.

4 Privacy in federated learning

Privacy in FL refers to the methods that guarantee the protection of sensitive data used by edge devices for training the local model during collaborative learning.

4.1 Attacks in federated learning

In traditional machine learning, attacks focus on centralized datasets. However, vulnerability in FL arises from the independence of clients, their potentially diverse data, and repetitive aggregation of updates. These characteristics may lead to several types of adversarial attacks aiming to manipulate the learning outcome, extract sensitive information, or degrade the final global model. FL attacks can be broadly classified into two types: privacy attacks and integrity attacks. Privacy attacks seek to compromise the confidentiality of client data, while integrity attacks attempt to disrupt the learning process or manipulate the final global model. The various types of privacy and integrity attacks are shown in Fig. 7.

4.1.1 Privacy attacks

The privacy attack can be classified into inference attacks and reconstruction attacks based on the quantity and precision of data the attacker is targeting. Privacy attacks in FL can be understood from the mathematical formulation of the training process. As described in Sect. 3.1, each client minimizes the local loss function using its private

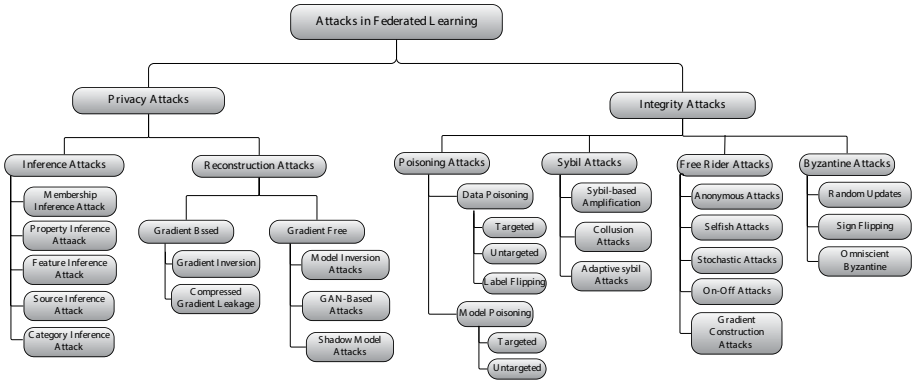


Fig. 7 Taxonomy of attacks in Federated Learning, classifying threats into privacy attacks and integrity attacks along with their key subtypes

dataset, and the corresponding gradients or model updates are shared with the central server for aggregation using FedAvg as in Eq. 2. Since these updates are computed as functions of the underlying data samples, they inherently encode sensitive information about the client's dataset. Consequently, an adversary observing these updates can exploit this dependency to infer hidden properties of the data or reconstruct the original inputs. This establishes the mathematical basis for privacy attacks in FL.

4.1.1.1 Inference attacks Inference attacks aim to reveal sensitive information about the client data. The iterative exchange of the model updates between the client and the central server can leak information about the underlying data. Attackers could exploit this vulnerability to determine specific data points or attributes. Inference attacks can be divided into membership inference attack, property inference attack, feature inference attack, source inference attack, and category inference attack, depending on the type of information the attacker is targeting. The Membership inference attack (MIA) determines whether a particular data point is used by the client during its local training. Attackers use the difference in the model updates to check whether a data point was present in the dataset. In overfitting-based MIA [36], the server will create an overfitted model to make the model memorize the training samples. The model is then queried with a candidate sample and infers membership if the model shows high confidence, as memorized samples tend to yield lower entropy. A perfect membership inference attack with 100% confidence is demonstrated in [37] when the server is dishonest. The Property inference attack aims to infer the properties of the entire training data, which is not intended to be exposed by the global model. In [38], a property inference attack is performed by training an auxiliary classifier that maps the gradients of the target client to property labels. The classifier is used to infer statistical characteristics of a client's private dataset, even if those properties are not related to the training task. The objective of the feature inference attack is to determine the characteristics of an individual data point within the dataset. In [39], the attacker guesses the missing features of a single data point by repeatedly sending inputs to the vertical FL prediction system and adjusting their guesses until the output matches what they observed. This allows the attacker to accurately identify the private feature values without seeing any training data or model details. In source inference attack [40, 41], the adversaries aim to identify the origin of a particular training record. In category inference attack [42], the adversary is attempting

to seek the category of the client's data based on the model updates shared between the client and central server during the model training.

4.1.1.2 Reconstruction attacks Another class of privacy attacks, called reconstruction attacks, attempt to reconstruct the client data or the representation of the client data from shared model parameters. Reconstruction attacks can be broadly classified into gradient based and gradient free methods depending on the attacker's access to the shared gradients. In gradient based reconstruction attack, attackers use the gradients or the model parameters transmitted by the individual clients to reconstruct the training data or its features. In contrast, gradient free reconstruction attack considers the model as a black box and uses the model output to reconstruct the data.

Gradient inversion attack utilizes the raw gradients transmitted during training to reconstruct the input data. The initial attempt to reconstruct training data from gradients was proposed in [43]. In Deep Leakage from Gradients (DLG), the attacker initializes dummy data and labels randomly. The dummy data is then iteratively adjusted until its gradients match the gradients from the original data. Improved DLG (iDLG) [44] uses the mathematical insight that the gradient signs on the output layer of cross entropy classification directly reveal the true label. As the true label is already known, iDLG utilizes DLG to reconstruct the features, resulting in faster convergence. Wasserstein distance-based DLG (WDLG) [45] replaces the standard loss function in DLG with the wasserstein distance to match the dummy and real gradients, achieving faster convergence and higher fidelity than DLG.

DLG and its variants struggle to reconstruct original data when gradients are subjected to compression, as the distortion of gradient information significantly hampers the matching process. Although iDLG and WDLG outperform DLG in terms of convergence speed and reconstruction quality, all these methods remain sensitive to gradient compression, which partially mitigates gradient leakage while motivating the development of specialized compressed-gradient reconstruction attacks. Highly compressed gradient leakage attack (HCGLA) [46] is designed to work against highly compressed gradients, which are used to enhance communication efficiency in FL. Table 4 compares gradient inversion attacks in terms of reconstruction quality, convergence speed, and robustness to gradient compression.

Gradient free reconstruction attack operates without accessing the gradients and relies on black-box queries, surrogate models, or generative priors. Model inversion attacks are one of the types of privacy attacks in which attackers attempt to infer sensitive data used for training the model by leveraging access to the model's output. Model inversion attacks are introduced in [47] and show that a black-box attack targeting a predictive model can reconstruct inputs belonging to specific output classes. They achieved this by finding input vectors that optimize the model's confidence score for a target class. Their

Table 4 Comparison of gradient inversion attacks illustrating their performance in terms of reconstruction quality, convergence speed, and robustness to gradient compression in federated learning settings

Method	Reconstruction quality	Convergence speed	Robustness to compression
DLG	Moderate	Slow	Low
iDLG	High	Faster	Low
WDLG	High	Faster	Low–Moderate (limited)
HCGLA	Moderate	Moderate	Moderate (designed for compression)

experiments, particularly in face classification models, demonstrated that even with only class probabilities, the attacker could reconstruct distinct faces corresponding to the known class labels, posing serious privacy issues. The attack assumed that the adversary would have access to some prior knowledge, such as the target label and some surrogate data obtained from the same distribution. In [48], model inversion is investigated in cooperative inference scenarios, where various parties hold different components of a model. Partial access to model outputs or activations is shown to be adequate for successful inversion.

Generative Adversarial Networks (GANs) are deep generative models with two parts: a generator that attempts to produce realistic data and a discriminator that tries to identify real versus fake samples. In FL, the GAN-based attack leverages this setup to reconstruct personal training data possessed by different clients, usually without access to raw data or model parameters. A robust attack framework termed Generative Model Inversion (GMI) is proposed in [49]. It demonstrates how a pretrained generative model, such as GANs, can be used to reconstruct private training data from a target model without gaining access to its internal parameters or gradients. The process involves looking through a pretrained GAN's latent space to identify an input 'z' such that the target classifier generates high-confidence predictions from the generated image G(z). It is a gradient-free, black-box attack since the optimization is only directed by output confidence scores. Notably, utilizing solely the predicted class probabilities, GMI can reconstruct identity-preserving facial images that closely resemble the actual individuals in the training dataset.

Shadow model-based reconstruction attacks target the retrieval of sensitive data through the training of a shadow model imitating the global model's behavior. They typically arise when an attacker, either a participating client or an external observer, has access to partial input features and their corresponding model outputs. With this information, the attacker trains a shadow model that acts as an estimation of the decision boundary of the true FL model. After training, the shadow model is utilized to infer or reconstruct missing features of the data from other participants. For instance, in vertical FL, an adversary party can employ its feature set along with the labels to train an adversary shadow model. Subsequently, leveraging optimization or generative methods, the attacker can recover the remaining features of the same data point consequently compromising feature privacy. The attacker in [48] uses the activations and their respective output labels to train a GAN that serves as a surrogate or shadow model to estimate the inverse mapping from activations back to the input data. By gathering activation label pairs and employing them for offline training of the GAN, the adversary constructs a model that closely resembles the original input–output mapping, allowing it to reconstruct sensitive client inputs without having access to gradients or raw data. This attack is a shadow model reconstruction attack since the attacker learns a parallel model from visible data in order to derive private information, emphasizing an important privacy threat in collaborative and FL systems.

4.1.2 Integrity attacks

Integrity attacks in FL arise from the dependence of the global model on aggregated client updates. As defined in Sect. 3.1, the global model is iteratively updated by combining local model parameters or gradients received from participating clients through

aggregation mechanisms such as FedAvg (Eq. 2). Since the aggregation process assumes that client updates are reliable and aligned with the true optimization objective, any malicious manipulation of local updates can directly affect the global model. From a mathematical perspective, integrity attacks aim to corrupt the optimization process by injecting adversarial or misleading updates, thereby degrading convergence, reducing model accuracy, or inducing targeted misbehavior. It is categorized into four types: Poisoning attacks, Sybil attacks, Free-rider attacks, and Byzantine attacks.

4.1.2.1 Poisoning attacks Poisoning attacks in FL occur when the adversarial participants purposefully tamper local model parameters or training data to corrupt the training process and degrade the global model's performance. The two main categories of poisoning attacks are model poisoning and data poisoning [50].

In data poisoning attack [51, 52], the adversarial participants inject poisoned data into their local dataset, which may cause the global model's performance to deteriorate considerably. These attacks can be targeted, untargeted, and label-flipping attacks. Targeted poisoning attacks [53] manipulate the local training data to induce hidden behavior in the global model. Malicious entities inject poisonous data into their local datasets to manipulate the global model's behavior on specific input. These attacks significantly reduce the classification accuracy for the targeted class. Untargeted poisoning attacks [54] degrade the overall performance of the global model without giving significant importance to a specific data point. The objective is to prevent the model from converging or diminishing its overall efficacy. In the label flipping attack [55], the adversary changes the labels of certain data points in their local dataset to incorrect values. This type of attack targets classification models, where the integrity of the label is most important. Label flipping can be static or dynamic. Static label flipping [56] is a consistent method in which the labels of specific data points are constantly interchanged between classes throughout the entire training process. Dynamic label flipping [57] involves changing the strategy of label flipping during the training process, making detection and mitigation more difficult.

In a model poisoning attack [58], the model parameters obtained after 'e' epochs of local training are tampered before sharing them with the server. The principal objective of this attack is to compromise the global model. Targeted model poisoning attacks attempt to make the model misclassify specific inputs while keeping overall performance intact on non-targeted classes. Untargeted model poisoning attacks [59] attempt to degrade the global model, effectively reducing accuracy or reliability.

4.1.2.2 Sybil attacks Sybil attacks [60] involve a malicious entity that creates multiple fake identities to gain uncontrolled influence over the model training process, compromising the integrity of the training. Sybil attacks can be broadly classified into amplification based sybil attack, collusion based sybil attack, and adaptive sybil attack. Amplification based sybil attacks are a powerful type of integrity threat in FL, where a single adversary manipulates several fake or compromised clients, also known as sybil nodes, to increase the impact of malicious updates on the global model. The fundamental idea behind this attack is to exploit the proportion that the server assigns to each client update during global model aggregation. By adding a large number of sybil clients, an attacker can significantly affect the aggregated model update. The Sybil amplification attack was

initially presented in [61], demonstrating that the quantity of sybil clients significantly influences the effectiveness of label-flipping attacks in FL. They showed in one experiment on the MNIST dataset that an attack that tried to change digit "1" to "7" with just one malicious client was not successful. The attack worked with more than 96% accuracy when two sybil clients were added to outvote the one honest client holding digit "1". This shows that sybil amplification can be dangerous even with a relatively small number of nodes. In a collusion based sybil attack [62], sybil nodes collaboratively coordinate their malicious activities, therefore enhancing the probability of changing the global model. Collusion-based sybil attacks emphasize quality and synchronization of updates to bypass FL defense mechanisms, especially those based on client similarity. In an adaptive sybil attack [61], the sybil nodes dynamically modify their behavior according to the training state and detection mechanism. When anomaly detection algorithms are used, sybil nodes may reduce the effect of their changes or act only in specific training rounds to avoid detection.

4.1.2.3 Free rider attacks In FL, free rider attacks [63] is a type of integrity attack in which adversaries attempt to benefit from the final aggregated model without providing valuable information or computation during the training phase. Instead of developing a legitimate model on their local data, the malicious participants share fictitious, arbitrary, or stolen updates to obtain the improved aggregated model at each aggregation round.

Anonymous free rider attacks, selfish free rider attacks, stochastic free rider attacks, on-off free rider attacks, and gradient construction attacks are major types of free rider attacks in FL. Anonymous free rider attacks [64] attempt to benefit from the final model without contributing any data or computation resources during the training process. These attackers remain undetected by obscuring their lack of involvement, frequently by sending out irregular or outdated model updates. Selfish free rider attackers [64] contribute insignificantly during training while gaining benefits from the global model. Although these attackers may occasionally offer legitimate updates to avoid detection, their main objective is to use as minimal resources as possible while still obtaining access to the final aggregated model. In a stochastic free rider attack [63], the malicious participants use stochastic updates to disguise their lack of contribution. In on-off free rider attack [65], the malicious entity alternates between providing useful updates or malicious updates during various training rounds. Due to this intermittent behavior, the attacker can appear cooperative during certain rounds and blend in with genuine participants, making it harder for the system to identify malicious activities. In gradient construction attack [66], the attacker attempts to generate gradient updates without training data locally.

4.1.2.4 Byzantine attacks In FL, byzantine attacks are a type of adversarial threat in which one or more clients act maliciously or arbitrarily, intending to compromise the global model's performance or convergence. These clients, referred to as byzantine clients, can transmit arbitrary updates or modify the training process in unforeseen manners.

In random update attack [67], malicious clients send noisy or arbitrary gradients that lacks correlation to the local training data. In schemes like FedAvg, where the central server calculates a simple average of all updates received from the client, these updates

add significant variance to the aggregation process. The global model may diverge completely or converge very slowly as a result of the unpredictability of these adversarial updates. This is particularly problematic in Non-IID environments, where anomaly detection is already challenging due to the inherent diversity in client updates.

In sign flipping technique byzantine clients compute the correct gradient direction but purposefully flip the sign of the update before sending it to the server. Compared to pure random updates, this approach has a more structured disruptive effect and can substantially hinder convergence. Sign-flipping attacks can bypass a number of current robust aggregation mechanisms, including Krum, Median, and Trimmed Mean, as shown by [68]. Attackers can increase the disruption's effectiveness while staying statistically close to benign clients by manipulating the flipped gradient's magnitude or flipping specific components.

Omniscient byzantine attackers are fully aware of the FL training pipeline, including the server's aggregation rule, the global model state, and even the updates from trustworthy clients. By carefully crafting malicious updates that closely resemble honest gradients, attackers can use this privileged information to gradually guide the global model towards a malicious goal, like backdoor insertion or targeted performance degradation. The "Little is Enough" attack detailed in [69] involves a small group of colluding clients who strategically optimize their updates to effectively and privately compromise the model. To reduce detectability and enhance adversarial impact, these updates were generated through a constrained optimization problem.

The following points provide a concise summary of the primary insights obtained from the examination of attacks in FL:

- *Decentralization introduces new attack surfaces:* Unlike centralized learning, FL relies on independent clients and iterative aggregation, creating multiple entry points for adversaries. Malicious clients can exploit this decentralized structure to manipulate updates or extract sensitive information.
- *Privacy leakage in gradient sharing:* Even without access to raw data, shared model updates can leak sensitive information. Gradient-based attacks, such as reconstruction and inference attacks, demonstrate that private data can be recovered or inferred from model updates, exposing a fundamental privacy vulnerability.
- *Attack diversity increases system vulnerability:* FL is exposed to a wide range of attacks, including inference, reconstruction, poisoning, sybil, free-rider, and byzantine attacks. These attacks compromise both privacy and model integrity simultaneously.
- *Non-IID data amplifies attack effectiveness:* Heterogeneous data distributions across clients make it easier for adversaries to hide malicious behavior. Distinguishing between natural variation and adversarial manipulation becomes significantly more challenging.
- *Collusion based attacks:* Coordinated adversaries (e.g., sybil attacks and collusion-based poisoning) can significantly influence the global model. Even a small number of malicious clients can dominate the aggregation process.
- *Gradient manipulation as a critical attack vector:* Many attacks directly exploit gradients, including gradient inversion, sign-flipping, and model poisoning. Since gradients are the primary communication medium in FL, they represent a major vulnerability.

- *Effectiveness of black-box attacks:* Even without access to gradients or internal parameters, attackers can reconstruct sensitive data using model outputs (e.g., model inversion and GAN-based attacks). This indicates that limiting internal access alone is insufficient.
- *Adaptive and stealthy attacks:* Advanced adversaries dynamically modify their behavior (e.g., adaptive Sybil attacks and on-off free-rider attacks) to evade detection, making them difficult to identify using traditional defense mechanisms.

4.2 Attacks defense strategies

The choice of defense mechanisms for ensuring model integrity and privacy in FL depends on the specific characteristics of the underlying attacks.

4.2.1 Inference attack defense strategies

Inference attacks in FL involve attackers attempting to extract sensitive information from shared model updates. One of the defense approaches is to deploy a digestive neural network [70], which alters the input data to reduce the accuracy of inference assaults while maintaining high classification accuracy. The digestive neural network outperforms traditional DP algorithms, with higher classification accuracy and reduce attack success rates. Client-level Input Perturbation (CIP) [71], which shifts each client's local data distribution to create a personalized perturbation, is one of the effective defense mechanisms for membership inference attacks. HE and SMPC are the common defense mechanisms against feature inference attacks to defend the local models. However, these techniques can be computationally expensive. Another way is to use lightweight keys based on the computational diffie-hellman problem [72] to encrypt local models, ensuring that the server only receives aggregated model updates and has no access to individual models. For effective membership inference attack, defenses include the application of bias-based techniques, which concentrate on the bias parameters of neural networks rather than weights. This method maintains excellent inference accuracy while lowering computational overhead. Defenses against category inference attacks include the use of Confusional AutoEncoders (CoAE) [73], which distort the data in a way that makes it difficult for adversaries to infer the correct category. This method has been shown to effectively block label inference attacks without significantly impacting the accuracy.

4.2.2 Reconstruction attack defense strategy

In FL, reconstruction attacks take advantage of the shared gradients or model updates to deduce confidential information about the data of specific clients. The goal of defensive strategies against these attacks is to preserve model performance while obscuring or safeguarding the client information. Gradient inversion defense strategies include Gradient Sign Inversion (GSI) [74], which disrupts adversarial example generation by inverting gradient directions. Gradient Correction [75] adjusts backpropagated gradients to address issues like over-transmission in activation functions. The attacker must rely on input–output queries in black-box settings since they lack access to the model's internal parameters. To defend against attacks, adversarial samples are injected into GAN-based frameworks during training. To provide a strong defense against black-box attacks, techniques such as the approximate gradient sign method [76], which utilizes differential evolution and generative model inversion that change the decision space or

employ conditional generative models. The assumption behind grey-box attack is that the attacker knows only a portion of the model, such as gradients, but not all the parameters. In FL environments, methods such as batching and gradient mixing prevent grey-box attacks including external gradient inversion attacks [77]. The two main defenses against approximate gradient inversion attacks are GSI and analog computing with synthetic gradients [78], which uses hardware co-processors to mask gradient parameters through nonlinear random projections, making precise approximation more difficult. Furthermore, to inject fake features during training, GAN-based protection frameworks [79] use fictitious samples, which makes it more difficult for attackers to carry out trustworthy gradient inversion.

Privacy-encoded FL (PEFL) [80], which decomposes large weight matrices into multiple cascading sub-matrices, introducing an entangling nonlinear mapping between model gradients and raw data, is one of the defense strategies for DLG. Gradient Perturbation [81], adds noise to the gradients to obscure the information. Dynamic DP parameters, which use dynamic privacy parameters to inject adaptive noise variance aligned with the trend of gradient updates, are the other defense mechanisms for DLG. One of the defense strategies against HCGLA is redesigning the objective function [46], which ensures that compressed dummy gradients are like compressed real gradients. It also uses the Init-Generation Method [46], which initializes dummy data through a novel method to more effectively align with the compressed gradients. Against gradient leakage attacks, unbiased sampling-based secure aggregation serves as a baseline defense strategy. However, adversaries can utilize advanced refinement steps which narrow the gradient searching range, approximate the magnitude of each gradient value, and revise gradient signs to bypass this defense. To counter this vulnerability, [82] introduced an additional step to hide the sampled gradient indices, which provides an essential layer of protection to secure individual user updates.

In [83], the authors proposed ResSFL, a split FL (SFL) framework that is intended to withstand MIA while being trained. It is based on the use of attacker aware training to derive a resistant feature extractor that is then used to initialize the client side model before standard SFL training. A framework called Knowledge Distillation Resistance Transfer for Split Federated Learning (KDRSFL) [84], which combines one-shot distillation techniques with adjustment strategies optimized for attackers, is one of the other defense strategies for MIA. KDRSFL improves the accuracy of the classification of feature extractors and strengthens their resistance to adversarial attacks.

4.2.3 Poisoning attack defense strategy

Depending on the type of attack, different tactics are needed to defend against data poisoning. GAN-based mimic models [85] and diversified ensemble classifiers can detect and reduce suspicious samples for targeted data poisoning attacks. Influence-function-based data sanitization [86] is used to filter anomalous points and mixed-strategy game models to approximate Nash equilibria to fight untargeted data poisoning attempts [87]. Label-flipping attacks are identified with Local Intrinsic Dimensionality (LID) [88], and mislabeled data was de-emphasized in high-dimensional spaces. Targeted model poisoning attacks are mitigated using SVM classifiers [89] with feature engineering or ensemble-based data cleaning techniques to detect malicious inputs. For untargeted model poisoning attacks, influence functions in online learning setups, combined with

constrained optimization approaches [90] and robust data sanitization, reduce the negative effects of corrupted data on model performance.

4.2.4 Sybil attack defense strategy

Defending against sybil attacks requires a combination of strategies to defend against different adversaries. Methods like FoolsGold [61] and MAB-RL [91] are effective in defending against adaptive sybil attacks. SCA [62] looks for collusive behavior during aggregation, while FL-Defender [92] uses last-layer gradient analysis to reweight updates based on similarity to counteract collusion-based sybil assaults. To protect against gradient manipulation, ShieldFL [93] uses HE for gradient distance measurement, and CONTRA [94] employs cosine similarity and dynamic reputations for client evaluation. To mitigate label flipping, FL-Defender analyzes gradients, while minimum aggregation [95] reduces the impact by aggregating minimum update values. Most existing sybil defense mechanisms assume that malicious clients exhibit similar update patterns that can be detected through similarity analysis. However, adaptive sybil attacks dynamically modify their behavior to remain statistically close to benign clients, reducing the effectiveness of traditional similarity-based defenses in large-scale heterogeneous FL environments.

4.2.5 Free-rider attack defense strategy

In FL, defending against free rider attacks requires specific tactics for every kind of attack. While STD-DAGMM [66] detects anomalies in supplied gradients, methods such as the Parameter Audit-Based Secure and Fair FL Scheme(PASS) [96] safeguard accuracy and privacy against anonymous free rider attacks even in the presence of numerous adversaries. Model Weight evolving frequency defense [97] uses model weight evolving frequency to identify minimal contributors. PASS retains high success rates for selfish free rider attacks even in the presence of weight evolving frequency. Stochastic update detection examines update consistency in the event of stochastic free rider attacks [63], while the outpost defense mechanism uses selective noise addition to mask anomalous activity. Lastly, mixing gradient strategies prevents gradient building attacks, and PEFL [80] raises parameter complexity to discourage gradient fabrication.

4.2.6 Byzantine attack defense strategy

Byzantine attacks in FL are countered by defense strategies that protect the global model from malicious or manipulated client updates. The Krum algorithm, one of the defense strategies to prevent byzantine attack, was first introduced in [67]. By removing malicious updates during aggregation, krum seeks to protect against byzantine failures. Krum eliminates extreme or inconsistent updates by choosing one update that is closest to the majority in terms of euclidean distance rather than averaging all client gradients. This approach performs best when there are few byzantine clients and is predicated on the idea that most clients are honest.

In [98], an enhanced FL framework is presented that transcends conventional server only protection, where clients add a penalty loss to reduce divergence across benign models, and the server applies multiple learning and maximum clique detection to detect malicious updates. Through this defense, immunity to byzantine attacks is greatly enhanced, particularly in Non-IID scenarios, with malicious client identification. Although robust aggregation methods improve resilience against byzantine attacks,

Table 5 Comparison of defense strategies against inference and reconstruction attacks in FL, highlighting their privacy effectiveness and trade-offs in computation cost, communication overhead, and impact on model accuracy

Attack type	Method	Strong privacy	Comp. Cost	Comm. Cost	Accuracy impact
Inference	Digestive NN [70]	✓	High	Moderate	Low
	CIP [71]	✓	Moderate	Moderate	Low
	CoAE [73]	✓	High	Moderate	Low
Reconstruction	GSI [74]	✓	Low	Low	Moderate
	Gradient Correction [75]	✓	Moderate	Moderate	Moderate
	GAN-based Defense [79]	✓	High	High	Moderate
	PEFL [80]	✓	High	Moderate	Moderate
	Gradient Perturbation [81]	✓	Moderate	Moderate	Low

Table 6 Comparison of defense strategies against integrity attacks in FL, including poisoning, Sybil, free-rider, and Byzantine attacks, with respect to robustness, computational cost, scalability, and performance under non-IID data settings

Attack type	Method	Robustness level	Comp. Cost	Scalability	Non-IID robustness
Poisoning	GAN + Ensemble [85]	High	High	Moderate	Moderate
	Influence Functions [86]	High	High	Low	Moderate
	LID [88]	Moderate	Moderate	Moderate	Moderate
Sybil	FoolsGold [61]	High	Moderate	High	Moderate
	FL-Defender [92]	High	Moderate	Moderate	Moderate
	ShieldFL [93]	High	High	Moderate	High
Free-Rider	STD-DAGMM [66]	Moderate	Moderate	Moderate	Moderate
	PASS [96]	High	High	Moderate	High
	Weight Evolution Frequency (WEF) [97]	Low	Low	High	Low
Byzantine	Krum [67]	Moderate	Moderate	Moderate	Low
	Clique Detection	High	High	Moderate	High

their effectiveness decreases in highly Non-IID environments where benign updates may naturally appear as outliers.

The Tables 5 and 6 provide a comparative overview of defense strategies against privacy and integrity attacks in FL, highlighting their effectiveness and trade-offs. As seen in Table 5, enhanced privacy-preserving measures generally come with higher computational and communication overhead. Cryptographic and GAN-based countermeasures provide increased protection against reconstruction attacks. However, its scalability is limited in resource-constrained FL settings. On the other hand, perturbation-based techniques are usually more efficient, but they are generally less immune to adaptive inference attacks. Table 6 highlights that defense effectiveness strongly depends on both adversarial sophistication and data heterogeneity. Similarity-based and statistical filtering mechanisms perform effectively against isolated malicious clients, but their robustness decreases significantly against colluding and adaptive adversaries in highly Non-IID environments. The results also indicate that defenses achieving stronger robustness under adversarial settings generally introduce higher computational complexity and reduced scalability. The subsequent points encapsulate the principal insights derived from attack defense strategies in FL.

- *Robustness vs. Efficiency trade-off:* Security mechanisms that provide strong guarantees, particularly those based on cryptographic techniques and adversarial

training, incur significant computational and communication overhead. This limits their scalability in real-world large-scale FL systems.

- *Lack of robustness in lightweight defenses:* Efficient defense mechanisms, such as statistical anomaly detection and simple aggregation strategies, offer improved scalability. However, they often fail against sophisticated and adaptive adversarial attacks, leading to reduced robustness.
- *Effect of Non-IID data distribution:* Most existing defense methods assume IID data across clients. In practical scenarios, data is inherently Non-IID, making it challenging to distinguish between benign heterogeneity and malicious behavior, thereby degrading defense performance.
- *Privacy vs. Security conflict:* Privacy-preserving techniques, such as DP and secure aggregation, obscure client-level information. While enhancing privacy, they simultaneously hinder the detection of malicious updates and anomalous behavior.
- *Absence of integrated defense mechanisms:* Current approaches predominantly focus on specific attack vectors (e.g., poisoning, Sybil, or inference attacks). There is a lack of unified frameworks capable of addressing multiple security threats simultaneously in FL environments.

4.3 Privacy protection techniques in federated learning

FL is a decentralized machine learning technique that maintains the locality of the underlying data while allowing models to be trained cooperatively across several devices or organizations. Although this method improves privacy by design, it is not completely resistant to complex attacks. Strong privacy protection strategies are therefore required to guarantee safe and reliable FL systems. Privacy protection in FL targets safeguarding the sensitive information during the training process. Techniques such as DP, HE, SMPC, federated averaging with secure aggregation, blockchain technology, and hybrid approaches play a vital role in enhancing privacy protection in FL systems.

4.3.1 Differential privacy

DP is the most widely used privacy protection technique in FL. The main idea is to make sure that users involvement in the training process has no particular impact on the final result, thereby making it difficult to infer specific information about the user data. This can be achieved by introducing noise to the data or the learning parameters, ensuring a trade-off between privacy and utility.

In mathematical terms, for any two datasets D and D' a randomized mechanism $\mathcal{M}$ adheres to (ϵ, δ), where ϵ denotes the privacy loss quantifying privacy levels, and δ indicates the probability that the mechanism fails to uphold the ϵ-bound. (ϵ, δ) is a DP mechanism where two datasets differ in exactly one data point and for any possible output **S** of the mechanism as shown in Eq. 15.

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(D') \in S] + \delta \tag{15}$$

The probability that $\mathcal{M}$ applied to dataset D and a neighboring dataset D' produces an output in S differs by at most a multiplicative factor of e^{ϵ} plus a small additive slack δ.

DP can be categorized into Local DP(LDP) and Global DP(GDP). These methods are designed to protect the sensitive information from being inferred from the model updates. Before sharing data or model updates with the server, LDP involves introducing

noise on the client side of the process. This approach guarantees that the individual data or model updates are protected, making it difficult for adversaries to infer sensitive information. LDP is particularly useful in situations where the server cannot be completely trusted and it is more suitable against various attacks, such as membership inference attack and model inversion attack. To achieve LDP, a client computes its local update w_k and modifies it by adding noise ξ to obtain result as shown in Eq. 16 [99].

$$\tilde{w}_k = w_k + \xi, \quad \xi \sim \mathcal{N}(0, \sigma^2) \tag{16}$$

where $\mathcal{N}(0, \sigma^2)$ is the Gaussian noise with variance σ^2.

GDP involves adding noise to the aggregated updates shared by the clients at the server side. This approach guarantees privacy for the entire dataset by hiding the contribution of individual data points. GDP is useful in situations where the server is assumed to be honest or semi-honest and the concern is intended to protect the aggregated data rather than the individual data points. Let

$$D_G = \frac{1}{K} \sum_{i=1}^{N} \tilde{w}_k \tag{17}$$

represent the aggregated update from K clients [99]. The server perturbs D_G by adding noise:

$$\tilde{D_G} = D_G + \xi_s, \quad \xi_s \sim \mathcal{N}(0, \sigma^2) \tag{18}$$

A client-level DP mechanism, which aims to protect the entire client's dataset is proposed in [100]. This approach uses Gaussian noise to add noise to the aggregated model and random subsampling to select a subset of clients in each communication round. It also introduced a method to dynamically adjust the noise and to select the number of participants in each round to optimize the trade-off between privacy and model utility. A novel Local Central DP(LCDP) framework, which enhances the privacy in FL was proposed in [101]. The LCDP method combines the LDP and Central DP (CDP) that ensure data privacy by perturbing original data instances, gradients, and model parameters. The LDP focuses on protecting the data before sharing, while the CDP focuses on server side aggregation. The noise is added to both the input data and gradients during training, which satisfies (ϵ, δ)- DP. In [102], the authors address the integration of DP with personalized FL to protect client data while maintaining model performance. It presents Rényi DP-based PFL (RDP-PFL), a novel framework that guarantees data privacy throughout the meta-learning process and offers theoretical analysis on its convergence under assumptions of both convex and non-convex loss functions. A novel LDP-Fed framework [103] is introduced that integrates LDP to secure participants data during collaborative training. It uses advanced techniques to reduce the effect of noise while preserving competitive model accuracy, such as selective sharing of parameter updates and iterative privacy accounting.

Although DP provides scalable privacy protection with relatively low communication overhead, its effectiveness strongly depends on the privacy budget and data distribution across clients. DP-based approaches are generally preferable in large-scale FL systems where lightweight computation is required. However, under highly Non-IID settings, excessive noise injection can significantly degrade convergence and model accuracy.

Furthermore, strong DP guarantees may unintentionally reduce the effectiveness of anomaly detection and robust aggregation methods because malicious updates become statistically similar to noisy benign updates.

4.3.2 Homomorphic encryption

HE is another privacy preserving technique that can be used in FL to protect the sensitive information from adverse attacks. HE is a significant advancement in cryptographic techniques that addresses the challenge of performing computations on encrypted data. HE allows computations to be carried out on encrypted data without decrypting it, maintaining the privacy and security throughout the process. HE schemes can be classified into Partially HE(PHE), Somewhat HE(SWHE) and Fully HE(FHE). The PHE scheme supports either addition or multiplication operations on the encrypted domain. Paillier cryptosystems and RSA support PHE schemes. SWHE supports both operations, but each operation, especially multiplication, increases the noise in the cipher text. As the number of operations increases, noise becomes too large, making decryption impossible. FHE supports both addition and multiplication without restrictions on the number of operations.

xMK-CKKS HE [104] utilizes an improved multi-key HE protocol to encrypt model updates with an aggregated public key, requiring collaboration for decryption. It prevents privacy leakage and is resistant to collusion, but the major limitation is that, the method requires collaboration among all devices for decryption. Selective HE is discussed in [105], which applies an encryption mask to a portion of model updates using a gradient-guided mechanism to select the mask. However, this method may not be fully protected against all types of attacks. Adaptive batch HE [106] is an adaptive batch encryption framework that clusters clients based on similarity and system capability. But the complexity in clustering and key size determination are the major drawbacks. Trusted Execution Environment(TEE) based multi key HE cryptosystem to enhance privacy and efficiency in FL was introduced in [107]. But the major limitation is that the method is completely based on TEE, which may not be available in all environments. An improved Paillier algorithm to encrypt gradients, enhancing training speed, was utilized in [108]. However, it maintains model accuracy with reduced computation overhead, but it is limited to partially HE.

Compared with DP-based approaches, HE provides significantly higher confidentiality guarantees, as the computations are performed directly on encrypted updates. This makes HE more suitable for highly sensitive sectors such as healthcare and finance where data leakage cannot be allowed. However, the encryption, decryption and key management processes add major computational and communication overheads, restricting scalability in large-scale cross-device FL settings with resource-constrained clients.

4.3.3 Secure multiparty computation

SMPC is a cryptographic technique that enhances privacy in FL by distributing a computation between multiple parties where no individual party can see the data of the other parties. By integrating SMPC into FL, it is possible to achieve a more robust privacy-preserving framework, ensuring that no single party can access the complete data of others, thus mitigating risks of data leakage. The mathematical formulation of SMPC is as follows: Let there be K clients or parties $c_1, c_2, \ldots, c_k$, each holding private

data D_k, where $k \in \{1, 2, \ldots, K\}$. The objective is to jointly compute a public function $f(D_1, D_2, \ldots, D_k)$ without revealing D_k to a third party. The function definition is $f : \mathbb{D}_1 \times \mathbb{D}_2 \times \cdots \times \mathbb{D}_k \to \mathbb{Y}$ where:

- $\mathbb{D}_k$ is the domain of inputs for party c_k,
- $\mathbb{Y}$ is the output space.

Each party c_k splits its input D_k into n shares $d_k^{(1)}, d_k^{(2)}, \ldots, d_k^{(n)}$ such that:

$$x_k = \text{Reconstruct}(d_k^{(1)}, d_k^{(2)}, \ldots, d_k^{(n)}),$$

where $d_k^{(j)}$ is sent to party c_j, ensuring no single party learns D_k entirely.

Parties collaboratively compute $f(D_1, D_2, \ldots, D_k)$ on the shared inputs without reconstructing the inputs during intermediate steps. The computation is performed as:

$$f(D_1, \ldots, D_k) = h\left(\sum_{i=1}^{n} d_1^{(i)}, \sum_{i=1}^{n} d_2^{(i)}, \ldots, \sum_{i=1}^{n} d_k^{(i)}\right),$$

where h operates on aggregated shares. After computation, the parties reconstruct the output y without learning any individual inputs:

$$y = \text{Reconstruct}(y^{(1)}, y^{(2)}, \ldots, y^{(k)}),$$

where $y^{(k)}$ is a partial result known to party c_k.

In [109],the authors introduce a secure aggregation scheme for FL using modified ElGamal encryption and super-increasing sequences to compress multi-dimensional data. The results show that it enhances privacy without relying on a trusted third party and reduces communication overhead. In [110], authors introduce an MPC-based framework to assess FL techniques against privacy inference and poisoning attacks, facilitating efficient FL systems. This method addresses both privacy inference and poisoning attacks in FL. In [111], the authors propose a decentralized FL algorithm using federated groups and SMPC to eliminate reliance on central servers, enhancing privacy and resistance to attacks by decentralizing FL. In [107], authors use multi-key HE to enable secure aggregation while preserving model accuracy in multi-party setups and it maintains accuracy while ensuring security, with reduced decryption complexity.

Approaches based on SMPC enhance privacy by guaranteeing that no single entity involved in collaborative computation obtains access to the whole client information. These strategies are useful especially in decentralized and trust-limited contexts. However, SMPC protocols usually involve numerous rounds of interaction and synchronization among the participants, which further increases the latency and the complexity of communication. The performance may further worsen in large-scale heterogeneous FL systems with intermittent client participation.

4.3.4 Hybrid approaches: combining techniques for enhanced privacy

In FL, the requirements of data privacy and model performance often stand in opposition to each other. There are various individual privacy preserving techniques such as DP, HE and SMPC that have been developed to address specific privacy concerns. But as a standalone solution, these strategies have severe drawbacks such as high computational cost and low accuracy. These disadvantages have resulted in the development of

new perspectives capable of addressing the strengths and weaknesses of various methods, such as the use of hybrid strategies that combine more than one privacy-enhancing technology.

4.3.4.1 Combining homomorphic encryption and differential privacy The combination of HE and DP attempts to achieve a multi-layered approach to privacy protection in FL. Each technique discussed unique aspects of privacy. HE protects the privacy of data during computation. This method makes it possible to act on encrypted data without decryption, while DP ensures the protection of learning processes. It does not allow the adversary to infer specific information about each data point, thereby safeguarding individual privacy and preventing unauthorized access to sensitive information. When combined, these techniques provide strong end-to-end privacy, which ensures that the data remains intact. Additional noise obscures anonymous inferences and any potential inferences arising from model updates. This combination provides a balance of privacy and randomness. HE and sensitive data transmission are done securely, while DP ensures anonymity on each sample. In addition, the integration of both approaches also improves threat mitigation, with HE preventing eavesdropping or malicious server behavior and DP ensuring that even malicious participants cannot extract sensitive details from the attacker.

In [112], the authors focus on the use of symmetric HE to defend against data leakage attacks in horizontal FL, providing an efficient encryption technique with reduced computational complexity and enhanced data privacy. A secure FL scheme employing DP and HE to protect data privacy during model training, enhance data security, maintaining model performance, and mitigating privacy risks in FL is presented in [113]. In [114], the combination of DP and HE in FL allows for less noise while preserving privacy with model accuracy. To integrate DP and HE in FL, a stochastic quantization operator was presented in [115]. This operator ensures DP guarantees with quantized noise, achieving a state-of-the-art privacy/utility trade-off with minimum computing overhead.

4.3.4.2 Combining secure multiparty computation and differential privacy In FL, a strong privacy-preserving technique is created that protects the privacy of individual data by combining SMPC with DP. SMPC prevents any third party from accessing the whole data, ensuring confidentiality during computation. DP provides mathematical guarantees against adversarial attacks by masking individual contributions. In [116], the authors present a privacy-preserving FL protocol for financial firms, enabling collaborative learning by using DP and SMPC without exposing sensitive client data, potentially improving models for optimal trade execution, credit origination, or fraud detection. In [117] a hybrid approach that combines DP and SMPC was introduced to balance privacy and accuracy in FL, outperforming state-of-the-art solutions.

Hybrid privacy-preserving approaches aim to mitigate the shortcomings of standalone strategies by combining mechanisms such as DP, HE, and SMPC. While these approaches improve overall security coverage, they bring additional system-level trade-offs. For instance, combining DP with safe aggregation or resilient aggregation algorithms might decrease the detection of poison and byzantine attacks, as the noise added can cover the harmful update patterns. Likewise, the combination of cryptographic algorithms with adaptive defense measures introduces a higher computational complexity

and training latency. These data imply that more privacy protection may not always lead to improved overall robustness, especially in large-scale Non-IID federated setting.

4.4 Privacy-utility tradeoff

This trade-off represents a key issue in FL whereby enhancing one property negatively affects the other two properties. For example, although DP and cryptographic solutions provide good data protection for sensitive data, they also lead to some degree of noise or additional computations, resulting in the decline of accuracy and efficiency of FL. Recently, there has been research showing that privacy, utility, and efficiency in FL cannot all be optimized at the same time. Consequently, an inherent no-free-lunch problem exists with respect to these three parameters [118]. The reason is that noisy DP causes degradation of the model utility in Non-IID data.

In terms of security, improving robustness in response to adversarial attacks (such as poisoning attacks or inference attacks) may require certain modifications in terms of imposing constraints and anomaly detection or robust aggregation and will influence the use of models and system performance. Some recent studies have considered the optimization of such trade-offs by employing adaptive privacy techniques and effective noise injection [119]. But most existing approaches focus on solving such trade-off problems involving two aspects at the same time (like privacy and utility or privacy and communication).

Parallel to these developments, significant efforts have been made towards privacy-preserving FL. Recent studies investigate the privacy-utility trade-off using dynamic game-theoretic approaches [120], particularly in large-scale model fine-tuning scenarios. Social-aware clustered FL frameworks [121] enhance personalization while incorporating customized privacy mechanisms. Secure FL has also been explored in wireless environments such as UAV-assisted crowdsensing systems [122], ensuring robust and privacy-aware aggregation under dynamic conditions. Furthermore, verifiable and cryptographic-based schemes [123, 124] have been proposed to guarantee both data privacy and aggregation correctness, while improving communication efficiency. Recent works have also focused on enhancing robustness against adversarial threats, proposing privacy-preserving FL models [125] resilient to model poisoning attacks, as well as integrating explainability [126] and adaptive cyber-defense mechanisms [127] in IoT and cyber-physical-social systems.

Recent studies have further advanced privacy-preserving FL by addressing challenges related to system heterogeneity, robustness, and communication efficiency, as demonstrated in [128] and [129]. In this context, recent works such as HE-FLAD [130], VAE-based anomaly detection with unlearning [131] and TMRGNN [132] further contribute by integrating privacy-preserving mechanisms with robust malicious client detection and adaptive defense strategies, demonstrating improved resilience in adversarial and Non-IID environments. These works highlight the growing emphasis on balancing privacy, utility, and resilience, while also revealing persistent gaps in maintaining model performance under strong privacy constraints and dynamic adversarial settings.

5 Communication efficiency in federated learning

FL is a distributed approach for training machine learning models across multiple edge devices without sharing sensitive data with a central server. However, it introduces communication overhead due to the frequent transfer of model updates and global model parameters between the client and the server. Using large deep learning models with more parameters results in a noticeable increase in communication overhead. Other factors resulting in communication overhead are multiple rounds of communication, energy consumption on edge devices, and heterogeneous data distributions. This section presents a comprehensive taxonomy of the solutions proposed to handle the challenge, as well as a review of the current state of FL research on communication efficiency. The taxonomy of different strategies to improve FL is illustrated in Fig. 8.

5.1 Model compression

One of the solutions to reduce communication latency in FL is to reduce the number of bits to be transmitted in an FL round. Model compression methods such as quantization, pruning, and sparse updates play a critical role in compressing the size of updates to be transmitted in FL systems to improve communication efficiency without impairing model performance to a considerable extent.

Quantization-based compression reduces the communication bandwidth needed for FL by reducing the number of bits utilized to convey local model updates. It accomplishes this by converting high-precision numerical data into a lower-precision format, such as fixed-point or integer types. Quantization can be classified into *uniform quantization*, in which data points are uniformly distributed within a fixed range, and *non-uniform quantization*, which adapts encoding based on data distribution. Through the use of quantization, FL systems can send model updates more efficiently with acceptable levels of accuracy.

In FedPAQ [133], clients compress their local model changes to smaller quantized representations before transmitting them to the parameter server. This technique is based

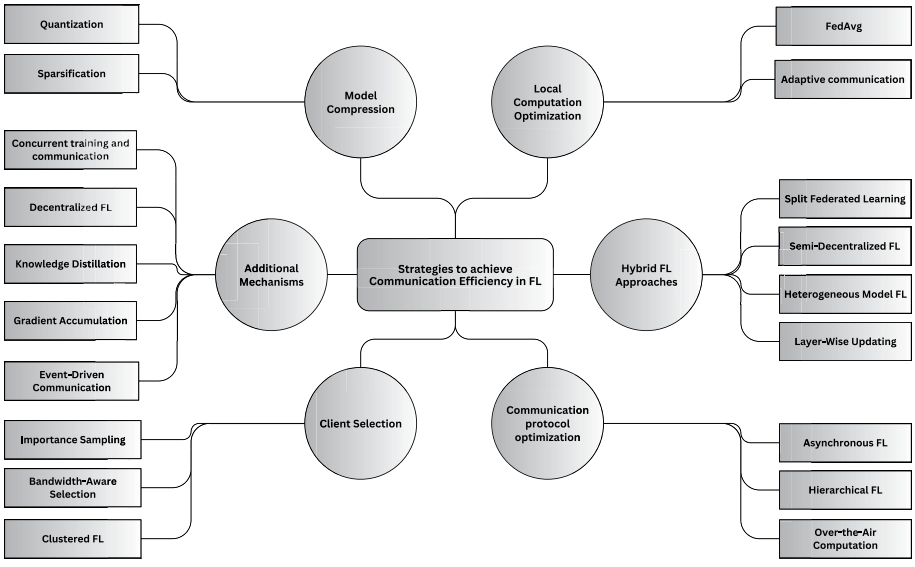


Fig. 8 Communication efficiency in FL: a taxonomy of strategies that minimize communication overhead through compression, local computation optimization, communication protocol optimization, client selection, and hybrid methods jointly reduce communication overhead and improve scalability

on quantized message passing. Communication overhead can be reduced by restricting the amount of data to be transmitted across the network. In order to counteract inefficient quantization effects and reduce average quantization errors, the Two-Layer Accumulated Quantized Compression Algorithm (TLAQC) [134] introduced RQSGD (Revised Quantized Stochastic Gradient Descent) with zero value correction. The method reduces the number of communication rounds while simultaneously reducing the cost of each round. To further save communication costs, TLAQC also uses adaptive thresholding and a parameter self-examination technique to decrease the gradient upload frequency. To reduce the loss of precision caused by compression, it additionally builds up quantization errors and stored weight deltas to compensate for the loss of gradient information.

TernGrad [135] applies ternary gradients that store values in the form of $\{-1, 0, 1\}$, thereby reducing data exchanges between clients and the server, consequently speeding up the training process. It also mathematically establishes the convergence of TernGrad under certain conditions and adopts methods like gradient clipping and layer-wise ternarization to boost convergence speed.

Sparsification reduces the number of non-zero elements in model updates, essentially compressing information being communicated during training. This is often done by truncating small magnitude gradients or weights to zero, thereby prioritizing communication for the most important parameters. Pruning, in contrast, eliminates redundant or less significant neurons and connections within neural networks, decreasing model size and communication overhead without compromising accuracy. Sparse updates focus on communicating only the most important updates rather than the complete model parameters to minimize communication overhead. The research [136], introduced a weight representation scheme for sparsely quantized neural networks, which is efficiently decoded during inference without affecting accuracy. AdaMC [137] employs additional networks to develop techniques for layer-wise sparsification and bias correction. It does this for IoT devices and maintains the accuracy of uncompressed FL. It is a framework that dynamically compresses model updates to fit the constrained resources of IoT devices, like bandwidth, energy, and processing power. It reduces transmission costs while maintaining model accuracy and allowing a wide range of edge devices to participate efficiently by changing compression algorithms during training.

FITCompress [138], presents a technique that combines layer-wise mixed-precision quantization and unstructured pruning using a single heuristic strategy, minimizing the trade-off between compression and model accuracy. Such techniques are usually applied in conjunction for the best compression results. An innovative framework in which networks are pruned and quantized during joint training with a significant reduction in size but slight decrease in performance was illustrated in [139].

A method that combines quantization with pruning to improve the representational power of binary networks with a few full-precision weights was proposed in [140]. FedDAC [141] is a communication-efficient approach that adaptively compresses model updates according to client-side resource variability and training dynamics. By changing compression algorithms across devices and rounds at the same time, it reduces communication costs while maintaining robustness and accuracy in heterogeneous FL situations.

5.2 Local computation optimization

Local computation optimization reduces communication overhead by reducing the frequency of data exchange at the cost of maintaining model precision. In FL, clients execute a number of local epochs 'e' before sending the local update to the server. This decreases the number of rounds of communication to achieve model convergence.

Federated averaging techniques are critical in optimizing the global model in FL because they balance communication economy, convergence speed, and robustness to heterogeneous data. In contrast to conventional gradient-based techniques, FedSGD [1] and FedAvg [1], are the foundational approaches that reduce communication overhead by training local models for several steps before averaging their parameters at the server, thereby reducing the number of communication rounds. Techniques like FedProx [142] limit local model divergence by introducing a proximal term, whereas SCAFFOLD [143] uses control variates to reduce client drift in order to handle the difficulties presented by Non-IID data. By using momentum-based updates, adaptive optimization techniques such as FedAdagrad, FedAdam and FedYogi [144] improve server-side aggregation and increase convergence stability. Different approaches, such as FedDC [145], use knowledge distillation (KD) rather than direct weight sharing, while communication-efficient approaches, such as FedNova, normalize heterogeneous local updates to avoid bias. These developments in federated averaging improve FL's performance and scalability, increasing its effectiveness in diverse real-world applications. Adaptive communication strategies further minimize communication costs by allowing clients to choose when to send local updates based on certain criteria, including the importance of local model updates or convergence measurements. FedLADA [146] utilizes a locally adaptive optimizer to improve FL efficiency and effectiveness by increasing training speed and preventing heterogeneous overfitting. It allows clients to perform more efficient local training, reducing the need for frequent communication with the server. As a result, FedLADA reduces the number of communication rounds required for model convergence, optimizing communication efficiency. Moreover, FedLADA utilizes global gradient descent to guide local updates, demonstrating strong empirical performance with improved accuracy and fewer communication rounds compared to baseline methods. Similarly, FedEAFO [147] is an efficient adaptive federated optimization algorithm that minimizes communication overhead by jointly considering two variables: local update and parameter compression. AdaptiveFL [148] is a bandwidth aware adaptive communication approach that adjusts the frequency and size of updates based on network performance. It also indirectly promotes bandwidth-aware client involvement by prioritizing clients with favorable communication circumstances. So it is primarily classified as an adaptive communication protocol, with a secondary role in client selection based on bandwidth.

5.3 Communication protocol optimization

Communication overhead can also be minimized by optimizing the communication protocol, the set of rules defining how the data are exchanged between devices. Asynchronous FL enables clients to communicate at varying time intervals rather than synchronized updates. This method speeds up convergence in heterogeneous networks by removing performance bottlenecks due to slow clients. In FedAsync [149] clients update their models locally and transmit the local updates to the server whenever they

are ready. This method allows the server to continuously update the global model without waiting for all clients to complete their local updates. Asynchronous techniques are prone to staleness, where updates are made against stale global models. Staleness refers to delayed client updates, which reduce model accuracy. The authors addressed this issue by using a mixing hyperparameter technique to avoid staleness caused by asynchronous updates.

Spyker [150] is an asynchronous multi-server FL system for geo-distributed clients. It employs a token-based approach to handle server interactions and avoid unnecessary model broadcasts. The servers pass a token to all the clients. The token aggregates the age of all server models, thereby facilitating the handling of asynchronous updates. Only the server with the token can trigger the transfer of server models. This prevents concurrent and redundant model broadcasts, reducing complexity and enhancing efficiency. Servers exchange and consolidate their models asynchronously in a peer-to-peer(P2P) fashion. This means each server is able to continually process client updates without having to wait on other servers. Servers keep track of the age of their models and adapt the contribution of updates based on this age. This helps to manage model staleness by preventing older updates from dominating the global model too much. Spyker mitigates the effect of aggressive clients using a local learning rate decay strategy. This prevents server models from being skewed towards client data distributions that produce updates frequently. FedFix [151] is a novel modification of FedAvg that maintains convergence stability in asynchronous contexts by averaging updates across fixed time periods. Furthermore, various algorithms, such as CNN Asynchronous Federated Optimization (CAFed) [152], have been proposed for specific applications to improve communication efficiency.

Hierarchical FL adds a middle aggregation layer to alleviate the load on the central server, thereby optimizing communication efficiency by minimizing the average latency between clients and servers. HiFlash [153] leverages a hierarchical structure in which client updates are first aggregated locally on edge servers before being sent to the cloud, drastically minimizing the amount of data transferred over wide-area networks (WANs) and reducing communication cost and latency. Furthermore, HiFlash employs deep reinforcement learning to adaptively control model staleness, and through dynamic staleness control adaptation, it maintains high model accuracy despite asynchronous updates from different clients. FedCDC [154] is a hierarchical clustered FL framework made for IoT setups with several types of devices. It initially organizes clients based on the similarity of their model update. Then, it trains them within each group utilizing a daisy-chain communication system, where models are transmitted from one device to the next and improved in order to reduce gradient variance and make learning from sparse data easier. Finally, proximal optimization is used for inter-cluster information sharing. This allows clusters to communicate with each other selectively, which reduces communication costs while increasing convergence stability and model accuracy in Non-IID circumstances.

Over-the-Air computation [155] is an innovative technology that unifies computation and communication operations in wireless networks. It uses the superposition characteristic of multiple-access channels to provide simultaneous data transfer and computation with much less latency and communication overhead. Devices send data

concurrently, and the receiver combines these signals to receive the intended computation, such as sums or averages.

5.4 Client selection

Client selection techniques in FL are critical for optimizing model convergence, reducing communication overhead, and increasing overall efficiency.

In FL, importance sampling is used to select samples that are important or useful to the learning process rather than uniform sampling. Importance sampling can be employed in FL at two levels: picking which agents to include in each round and selecting the data point from each agent to train with. In [156], client selection is based on their local loss, which reflects how much they can enhance the global model. In [157], a performance metric is used to examine the effects of importance sampling and develop methods for non-uniformly sampled agents and data. They also demonstrated that data variability at each agent and model variability across agents are the two parameters that control performance in sampling without replacement. GreedyFed [158] improves FL's effectiveness by selecting clients based on their contribution to the model and using the Shapley Value as a criterion. Shapley Value determines how much value each client contributes to the model when included within various subsets of clients. This approach selects the most contributing clients in each round of communication, resulting in faster convergence and higher model accuracy. For the sake of tractability of the computation of Shapley Values for the large number of clients, GreedyFed uses a quick Monte Carlo approximation algorithm, GTG-SHAPLEY12.

Bandwidth-Aware selection is used to select clients according to their network conditions to prevent stragglers and achieve timely updates. Client selection with improved network connectivity can be used to prevent model update delays and ensure that the training process remains optimal. Context-aware Online Client Selection (COCS) [159] combines contextual data about clients, including computational power, network states, and geographic locations. The comprehensive strategy facilitates the selection of clients that can be useful contributors to the model update without causing delays and communication overhead. This strategy is formulated to respond in real time to evolving client conditions so that the most appropriate clients are chosen for each communication round. Responsiveness is important in dynamic environments where client availability and network conditions can change. AdaptiveFL [148] is a communication-aware FL framework that changes the number of clients and how often they are provided with updates depending on the amount of bandwidth available. The method reduces communication overhead while keeping convergence constant by adapting communication resources and letting clients send updates based on network limitations. Clustered FL involves clustering similar clients to reduce communication overhead and enhance convergence through the exploitation of data similarities. Clients are clustered according to their data distributions or model similarities so that more efficient updates and less variance in weighted aggregates can be achieved. The authors of [160] proposed a clustered sampling technique to cluster clients based on sample size or model similarity. The objective is to improve client representativeness and minimize the variance of stochastic aggregation weights in FL. Sample-size-based clustered sampling requires sorting clients according to their sample sizes and then allocating them into clusters based on these sizes. Clustered sampling by model similarity divides clients into groups based on how

similar their gradients are using hierarchical clustering methods (such as the Ward algorithm). Grouping similar clients reduces variance in aggregation weights, resulting in more stable model updates, thereby reducing communication overhead.

5.5 Hybrid federated learning approaches

Hybrid FL approaches aim to reduce communication overhead while preserving the performance of the models by merging multiple methods. These methods are vital in situations where data privacy and resource limitations are major concerns.

Split Learning (SL) divides a machine learning model into parts that are co-trained by different parties. SL is advantageous when devices have limited computational capacity, as it enables them to outsource heavy computation to more powerful nodes. The application of FL and SL together can be used to combine the benefits of both approaches. FL preserves privacy by localizing data, while SL facilitates effective model training by offloading computational workloads to devices with different capabilities. SplitFed [161] is a method that splits the model between the server and the clients. Rather than sending full model updates, only intermediate activations are exchanged. This reduces communication overhead by limiting the volume of data transferred between clients and the server. The methods to accelerate split FL in wireless networks were discussed in [162]. It utilizes the parallel training feature of FL and the privacy advantage of SL to enhance efficiency across wireless networks.

Semi-Decentralized FL uses multiple local aggregators rather than one global server. This design reduces communication with the central coordinator by letting local aggregators process model updates from nearby clients. FedGosp [163] combines gossip protocols with FL to provide improved data privacy and minimize the dependence on a central server. This is especially useful in situations where heterogeneity in the data is high and a central server could be a bottleneck or vulnerable to privacy threats. Utilization of a clustering algorithm aids in partitioning similar data points or clients, which can make model updates more efficient and predictions more accurate in heterogeneous settings. The decentralized approach of FedGosp can result in cost-effectiveness by minimizing the need for a high-performance central server and reducing the communication overhead.

Heterogeneous Model FL is an extension of FL that supports varied client models and enables them to learn models of different sizes according to their processing powers. It is vital in systems where devices differ in terms of hardware, software, and data. HeteroFL [164] supports the training of heterogeneous local models with different computation complexities. This capability allows clients with different capabilities to cooperate efficiently in the training process. It improves FL by adaptively allocating subnets based on each client's capability. This method makes the framework both computation and communication efficient. The system minimizes communication overhead by supporting asynchronous communication and sending only the necessary model updates. It is especially useful in situations where communication bandwidth is scarce or costly.

Layer-wise updating in FL enables clients to transmit only particular layers of the model rather than full model weights to the central server in every round. This enhances communication efficiency, as it sends much less data between clients and the central server. FedLWS [165] improves the generalization capacity of the global model by automatically modifying the shrinking factors for model weights across individual

layers. Unlike common shrinking factor-based approaches that shrink all model weights simultaneously, FedLWS computes the shrinking factors specific to individual layers. The shrinkage factor is calculated dynamically using the global model parameters. The technique starts with observing the factors that contribute to the shrinking factor when training the network. It then calculates the shrinking factor for each layer based on its distinctive features in the global model. Through layer-specific adjustments, FedLWS is able to optimize model aggregation with minimal data exchange. This makes more accurate model updates and could reduce the amount of data transmitted for aggregation.

The approach proposed by [166] computes the divergence between the global model from the previous round and each client's local model. A measure of each client's model deviation from the global model is provided by divergence. Then the most divergent layers of the client model are only transmitted and aggregated. By transferring only a smaller amount of data, it reduces the overhead of communication. The layers of a deep neural network are categorized into shallow and deep layers in [167]. The parameters of the deep layers are updated less frequently than those of the shallow layers, which helps reduce the communication overhead by minimizing the number of updates that need to be transmitted between clients and the server. This method involves aggregating local models on the server using temporal weights. The temporal weights are assigned on the basis of the timing of the updates, allowing the server to make use of previously trained local models. This method improves the central model's accuracy and convergence by utilizing historical data.

5.6 Additional methods

By overlapping local training with communication, Overlap-FedAvg [168] reduces communication latency and overall training efficiency by a large extent. Overlap-FedAvg improves FL efficiency by parallelizing model training and communication. It uses a hierarchical computing strategy for efficient use of resources, staleness compensation Mechanism to avoid inconsistencies, and the Nesterov Accelerated Gradient (NAG) algorithm for faster convergence. It also supports compression methods to further reduce the communication expense.

Decentralized FL is another option to the centralized FL framework, where clients communicate directly to each other instead of communicating to the central server. This framework improves communication efficiency by eliminating a bottleneck on a central server and avoiding redundant data exchange. In decentralized FL, clients share P2P updates, transferring model parameters or gradients with only some neighbors, commonly controlled by a network topology or a graph structure. The problem of communication bottlenecks in decentralized stochastic optimization, especially when the data is split among several machines that only communicate with their local neighbors, was discussed in [169]. The authors present CHOCO-SGD, a new gossip-based stochastic gradient descent algorithm that includes compressed communication to limit data transfer between nodes. The paper also introduces CHOCO-GOSSIP, a gossip algorithm for the average consensus problem. This algorithm has arbitrary compressed messages and achieves linear convergence.

KD in FL is a method to transfer knowledge from several distributed models or clients to the global model while minimizing communication overhead and accommodating heterogeneous client models. Rather than aggregating model weights directly as in

FedAvg, KD-based FL methods communicate knowledge in the shape of soft labels or logits, feature representations, or simulated data, which makes the process more communication-efficient and robust to non-identical client architectures. The mechanisms for integrating KD in FL systems are explained in [170] and [171].

Gradient accumulation improves communication efficiency by making updates less frequent, thus requiring less bandwidth [1]. Rather than sending gradients after each mini-batch, clients collect them across multiple local iterations and then send updates to the server. This decreases the number of communication rounds, thereby preserving bandwidth. This is particularly helpful for large models, where updating regularly overwhelms the network bandwidth.

Event-driven communication in FL is a strategy for improving communication efficiency by reducing unnecessary data exchanges between clients and the central server. Rather than requiring periodic or synchronous updates, event-driven communication allows clients to send updates only when predetermined conditions or events occur. This strategy saves a lot of communication overhead and results in greater scalability and reduced convergence time. In [172], a new framework for improving communication efficiency in vertical FL with complementary feature sets for common objects was introduced. In contrast to conventional synchronous vertical FL solutions, this event-driven solution wakes up only a portion of the clients depending on the availability of relevant data to minimize communication and computation expenses. The framework applies dynamic local Regret to address non-convex models in non-stationary scenarios to maintain steady performance under real-world settings. Theoretical analysis and experimental results demonstrate improved learning efficiency, decreases overhead, and preserves model accuracy, resulting in an efficient solution for asynchronous and dynamic data environments. A multi-server FL system called ConFederated Learning was introduced in [173]. ConFederated Learning introduced a stochastic gradient approach with a conditionally triggered user selection mechanism to efficiently minimize communication overhead. Experimentation demonstrates significant gains in the effectiveness of communication over existing methods, and theoretical analysis confirms a linear convergence rate.

Tables 7, 8 and 9 combined give a complete picture of how to improve communication in FL. The summary table breaks down different tactics into important groupings, showing their main principles, examples of their work, and the trade-offs that come with them. This helps you understand how a variety of techniques reduce communication overhead. Table 8 complements this overview by providing a comparative analysis of individual techniques in terms of their suitable deployment scenarios, key advantages, and practical limitations. Unlike the taxonomy-focused summary, this table highlights the trade-offs associated with different approaches and illustrates how their effectiveness varies under different system constraints, network conditions, and heterogeneous FL settings. Such a comparison provides deeper insight into the circumstances under which specific communication-efficient strategies are more appropriate and the challenges they may encounter in real-world deployments. The high-level comparison table, on the other hand, groups various strategies into larger groups and rates them on critical system-level factors, including how well they reduce communication, how much they cost to run, how well they converge, how well they scale out, and how difficult they are to understand.

Table 7 Summary of communication efficient techniques in FL

Category	Technique	Key idea	Representative works
Model Compression	Quantization	Reduce precision of model updates	[133–135]
	Sparsification	Transmit only important gradients	[136, 137]
	Pruning	Remove redundant parameters	[139]
	Hybrid Compression	Combine pruning + quantization	[138, 140, 141]
	Error Compensation	Accumulate quantization errors	[134]
Local Computation Optimization	Multiple Local Updates	Reduce communication rounds (FedAvg)	[1]
	Proximal Optimization	Limit client drift (FedProx)	[142]
	Control Variates	Correct gradient drift (SCAFFOLD)	[143]
	Adaptive Optimization	Improve convergence (FedAdam, FedYogi)	[144]
	Adaptive Communication	Send updates selectively	[146–148]
Communication Protocol Optimization	Asynchronous FL	Update without waiting for all clients	[149, 151]
	Multi-server FL	Distributed aggregation (Spyker)	[150]
	Hierarchical FL	Multi-level aggregation (HiFlash)	[153, 154]
	Over-the-Air Computation	Simultaneous communication + aggregation	[155]
	Application-specific Optimization	Task-aware communication (CAFed)	[152]
Client Selection	Importance Sampling	Select high-impact clients	[156, 157]
	Contribution-based Selection	Use Shapley value (GreedyFed)	[158]
	Resource-aware Selection	Select based on bandwidth/computation	[148, 159]
	Clustered FL	Group similar clients	[160]
Hybrid FL Approaches	Split Learning + FL	Share intermediate activations	[161, 162]
	Semi-decentralized FL	Local aggregators reduce load	[163]
	Heterogeneous FL	Different model sizes (HeteroFL)	[164]
	Layer-wise Updates	Transmit selected layers	[165–167]
	Temporal Aggregation	Use time-weighted updates	[167]
Additional Mechanisms	Overlapping Training	Parallel communication + computation	[168]
	Decentralized FL	Peer-to-peer communication	[169]
	Knowledge Distillation	Share logits instead of weights	[170, 171]
	Gradient Accumulation	Reduce update frequency	[1]
	Event-driven FL	Trigger-based communication	[172, 173]

A comprehensive examination of communication efficiency strategies in FL reveals several critical insights:

- *Trade-off between reducing communication and improving model performance:* Quantization, sparsification, and pruning are some of the techniques that can greatly minimize communication overhead. However, they also cause approximation errors that reduce the accuracy of the model. Effective approaches use error compensation or adaptive strategies to find the right balance between these two.
- *Local computing is a strong alternative to communication:* One of the best ways to reduce communication rounds is to increase the frequency of local updates, such as in FedAvg and its derivatives. But too much local computing might cause client drift, especially in Non-IID circumstances. To fix this, we need tools like FedProx and SCAFFOLD.
- *Data heterogeneity is still a major problem:* Non-IID data distributions have a big effect on how well communication efficient strategies work. Adaptive optimization and clustered FL are two examples of techniques that are specifically designed to reduce the problems that arise from having different types of data.

Table 8 Analytical comparison of communication efficient FL techniques

Technique	Best use case	Main strength	Main limitation/failure mode
Quantization	Bandwidth-constrained FL systems	Large reduction in communication payload	Low-bit quantization may amplify gradient distortion under highly Non-IID data
Sparsification	Over-parameterized deep models	Transmits only significant gradients	Repeated omission of important updates may slow convergence
Pruning	Resource-limited edge devices	Reduces model size and communication overhead	Aggressive pruning may reduce model capacity and accuracy
Hybrid Compression	Highly constrained communication environments	Balances multiple compression objectives	Accumulated approximation errors may destabilize training
Multiple Local Updates	Intermittently connected clients	Substantially reduces communication rounds	Excessive local training causes client drift
Proximal Optimization	Strongly Non-IID environments	Improves convergence stability	Requires careful tuning of proximal parameters
Control Variates	Heterogeneous client distributions	Mitigates gradient drift effectively	Additional optimization and memory overhead
Adaptive Communication	Dynamic wireless networks	Adjusts communication frequency adaptively	Adaptive policies may introduce instability
Asynchronous FL	Large-scale mobile FL systems	Eliminates waiting for straggling clients	Stale updates may bias convergence
Hierarchical FL	Edge-cloud deployments	Reduces long-range communication overhead	Cluster-level bias propagation may occur
Over-the-Air Computation	Wireless communication environments	Simultaneous communication and aggregation	Sensitive to channel noise and interference
Importance Sampling	Large client populations	Prioritizes informative updates	Sampling bias may reduce fairness
Contribution-based Selection	Accuracy-sensitive deployments	Improves convergence efficiency	Shapley-value estimation is computationally expensive
Clustered FL	Strongly Non-IID client distributions	Reduces aggregation variance	Cluster formation introduces coordination overhead
Split Learning + FL	Computation-limited devices	Offloads heavy computation	Intermediate activations may leak information
Heterogeneous FL	Diverse edge hardware settings	Supports different model capacities	Aggregation inconsistency across models
Layer-wise Updates	Deep neural network architectures	Reduces unnecessary parameter transmission	Layer synchronization complexity
Knowledge Distillation	Heterogeneous client architectures	Avoids direct parameter sharing	Indirect knowledge transfer may reduce accuracy
Event-driven FL	Dynamic real-world environments	Avoids unnecessary communication rounds	Poor trigger design may destabilize training

Table 9 High-level comparison of communication efficient strategies in FL

Category	Comm. reduction	Computation cost	Convergence impact	Scalability	Complexity
Model Compression	High (↓↓↓)	Low (↑)	Moderate (↓)	High (↑↑)	Medium (↑)
Local Computation Optimization	Medium (↓↓)	High (↑↑↑)	Mixed (±)	High (↑↑)	Medium (↑)
Communication Protocol Optimization	Medium–High (↓↓-↓↓↓)	Low (↑)	Sensitive (staleness)	Very High (↑↑↑)	High (↑↑)
Client Selection	Medium (↓↓)	Low (↑)	Positive (↑)	High (↑↑)	Medium (↑)
Hybrid FL Approaches	High (↓↓↓)	Medium (↑↑)	Positive (↑)	High (↑↑)	High (↑↑)
Additional Mechanisms	Medium–High (↓↓-↓↓↓)	Low–Medium (↑ − ↑↑)	Mixed (±)	High (↑↑)	Medium–High(↑ − ↑↑)

- *Asynchrony enhances efficiency but can also lead to staleness.* Asynchronous and semi-asynchronous FL methods reduce latency and optimize resource utilization, particularly in large systems. But stale updates can make convergence worse, emphasizing the need of staleness-aware weighting and adaptive learning rates.
- *Client Selection is very important for scalability:* Choosing a group of clients that are both informative and resourceful reduces unnecessary communication while keeping the quality of the model. But biased selection procedures could make the results unfair and less generalizable.
- *The best trade-offs come from hybrid approaches:* Using more than one strategy, like compression with adaptive communication or hierarchical aggregation, frequently makes communication more efficient than using a single method. Hybrid FL methods work best when there are different types of resources in the actual world.
- *Optimizing at the system level is just as important as optimizing at the algorithm level:* Over-the-air computation, hierarchical FL, and protocol optimization show that communication efficiency is not just a problem with algorithms but also a problem with system design.
- *Event-driven communication is a promising direction:* Instead of sending updates every now and then, using model changes or convergence criteria to trigger them can reduce unwanted transmissions by a lot, making it easier to scale in dynamic situations.
- *There is no one-size-fits-all answer:* The performance of communication-efficient strategies is contingent upon system limitations, including network circumstances, client availability, model size, and data distribution. So, choosing the right strategies needs a lot of thought about the deployment circumstances.

6 Optimizing privacy and communication efficiency

Several methods attempt to simultaneously enhance communication efficiency and privacy, but it is crucial to comprehend their effects on system robustness, as certain communication optimizations may compromise foundational security assumptions. Communication-efficient approaches are necessary for scalable FL, but if not correctly implemented, they can accidentally reduce byzantine robustness. Many robust aggregation methods, like the median, trimmed mean, and krum, assume that bad updates may be found as statistical outliers in the complete gradient space. But this assumption is broken by compression and sparsification, which reduce the statistical distinguishability of updates and lower their dimensionality. For example, if only a part of the gradient coordinates is sent, poisoned components that are not in the chosen subset stay hidden, and bad clients can change the sent coordinates in a way that makes them look like normal updates. Quantization also lowers the accuracy of gradients, which might hide subtle but harmful changes that are hard to determine apart from typical quantization noise.

Asynchronous FL makes this problem worse by adding staleness, which happens when updates are made on old global models. Adversaries can take advantage of this situation by making stale updates that are still adversarial and that don't pass consistency tests, which makes convergence worse over time. In decentralized federated learning, clients have limited communication with one another, complicating the overall visibility of the system's operations. This makes defense based on cross-validation less effective

and makes it easier for coordinated sybil attacks to happen. The findings demonstrate that communication-efficient techniques can fundamentally modify the assumptions underlying byzantine-robust aggregation, potentially diminishing the effectiveness of such defenses. So, when designing communication-efficient FL systems, one must consider both robustness guarantees and security. This section looks at cutting-edge techniques that improve both privacy and communication efficiency. The Table 10 gives the impact of communication method and privacy methods on accuracy and communication overhead.

In FedPC framework [174] each client has the freedom to choose their own hyperparameters, such as the number of local epochs, learning rate, batch size, etc. Additionally, clients send a ternary vector instead of the local model updates to the server after each FL round. The ternary vector illustrates how the model updates have changed during the last two epochs. Since the ternary vector provides a compressed representation of local model updates, communication efficiency will be achieved by reducing the amount of data that the client must send. The hyperparameters are distinct to each client, and the model updates are not communicated to the network, therefore, capturing the model parameters and inferring data about each client will be impractical, resulting in enhanced privacy. CRS-FL [175] involves transmitting only a subset of model updates

Table 10 Unified compatibility matrix of communication-efficient and privacy-preserving techniques in FL

Communication method	Privacy method	Comm. overhead	Accuracy impact	Compatibility/limitations
Quantization	Central DP	↓ payload, ↑ rounds	↓	Noise + quantization error compound; careful calibration required
	Secure Aggregation	↑	↓ (due to quantization)	Secure aggregation preserves accuracy; quantization controls degradation
	HE/SMPC	↑↑	↓ (due to quantization)	Encryption dominates cost; limited practical efficiency
Sparsification	Local DP	↓ payload, ↑ rounds	↓↓	Combined effect of gradient dropping and noise significantly reduces utility
	Secure Aggregation	↓	↓ (depends on sparsity level)	Sparse updates reduce communication but may affect convergence; SA does not impact accuracy
	HE/SMPC	↑	↓ (due to sparsification)	Sparse operations inefficient under encryption; limited scalability
Local Computation	Central DP	↓ rounds	↓	Client drift + DP noise degrade convergence, (FedAvg-style)
	Secure Aggregation	↓ rounds	≈	Efficient combination; SA preserves accuracy while local steps reduce communication
	HE/SMPC	↑	≈	High communication cost due to encrypted aggregation
Asynchronous FL	Central DP	↓ idle communication	↓	Staleness + DP noise reduce convergence stability
	Secure Aggregation	↑	≈	Difficult to synchronize SA in async settings; accuracy largely unaffected
	HE/SMPC	↑↑	≈	Encryption overhead limits scalability in asynchronous systems
Hierarchical FL	Central DP	↓ long-range communication	↓	Multi-level aggregation with noise accumulation impacts accuracy
	Secure Aggregation	↑	≈	Requires secure aggregation at each level; accuracy preserved
	HE/SMPC	↑↑	≈	High overhead due to repeated encrypted aggregation across layers

from each client to the server for aggregation. This selection of parameters is made using a stochastic coefficient based on poisson sampling to reduce the amount of data provided and increase the probability of obtaining unbiased zero-gradient updates. The minimal data transmission volume substantially diminishes the risk of privacy violations and communication overhead.

GFL-ALDPA [176] is a privacy preserving gradient compression based FL framework where each client compresses and exchanges gradients in every FL round for enhancing communication efficiency. It also dynamically allocates the privacy budget across communication rounds in order to minimize privacy budget and model noise. PCFed [177] addresses privacy and communication efficiency by using an adaptive sampling-based intermittent communication strategy for selecting the set of clients to participate in an FL round. This adaptive strategy ensures that the clients are selected in such a way that communication latency is reduced. While the privacy budget ϵ measures the degree of privacy to be ensured, it also accomplishes DP by introducing noise into the shared model updates. In [178], each client performs a two-stage gradient pruning process after calculating their local gradients and applies DP before sharing the local gradients with the server. In the first stage of pruning, gradients below a predefined threshold are pruned, and the second stage of pruning ensures that the most essential gradients are transmitted to achieve communication efficiency. Appropriate noise levels are determined and added to each selected gradient based on its significance for ensuring DP. The authors of [179] introduced a new FL paradigm that improves personalization, minimizes communication overhead, and maintains privacy. It proposes feature fusion-based mutual learning, in which clients share and fuse extracted feature representations rather than model updates, enabling personalized learning with minimal data transmission. The method also includes privacy-preserving mechanisms, including DP and secure aggregation, to maintain client data confidentiality.

We introduce a unified conceptual optimization view of FL to provide a principled understanding of the trade-offs between privacy, accuracy, and communication efficiency. In this view, the objective of an FL system can be interpreted as minimizing the total communication cost, typically characterized by the product of the number of communication rounds and the size of transmitted updates, subject to multiple constraints. These include (i) a target model accuracy requirement to ensure acceptable performance, (ii) privacy constraints defined by parameters such as DP guarantees (ϵ, δ), and (iii) system-level constraints such as limited computation, bandwidth, and partial client participation.

This formulation highlights the inherent coupling between these factors. For instance, stronger privacy guarantees (smaller ϵ and δ) require injecting more noise into model updates, which reduces their utility and slows convergence, thereby increasing the number of communication rounds. Similarly, aggressive communication reduction techniques such as compression or reduced participation may degrade model accuracy or stability, especially under heterogeneous and Non-IID data distributions. From this perspective, existing FL methods can be interpreted as optimizing different components of this unified objective: model compression reduces update size, local computation optimization reduces communication rounds, and client selection controls participation to balance efficiency and convergence. Consequently, communication-efficient and privacy-preserving FL should be viewed as a constrained optimization problem that

requires jointly balancing these competing objectives rather than optimizing any single factor in isolation. The interaction between communication-efficient and privacy-preserving techniques is summarized in Table 10, where the symbols ↓, ↑, and ≈ denote decrease, increase, and negligible impact, respectively, on communication overhead and model accuracy.

7 Effect of heterogeneity in communication and privacy

Heterogeneity is a primary challenge in FL that directly affects communication efficiency and privacy. In real-world FL implementations, heterogeneity comes from multiple sources, such as statistical heterogeneity (Non-IID data distributions), system heterogeneity (differences in processing power, energy, and bandwidth), and model heterogeneity (differences in model topologies between clients). These kinds of differences have a big effect on the number of communication rounds, the size and frequency of updates, and how FL systems converge as a whole.

In FL, communication efficiency is significantly influenced by statistical heterogeneity, which refers to the diverse distributions of client data. Under Non-IID conditions, local updates may diverge in their optimization trajectories, resulting in unpredictable gradients and a decelerated convergence. Consequently, additional communication rounds are necessary to get a specified precision in comparison to IID cases. Non-IID data can create client drift, which can degrade convergence and increase communication cost. To address the issue of unstable updates, methodologies such as FedProx [142] and SCAFFOLD [143] integrate correction mechanisms to enhance update stability. However, these frequently require more computing or transmission power. Additionally, substantial differences between updates obtained by each client can, in turn, cause privacy leakage [180], due to the uniqueness of each distribution and possible bias. Furthermore, heterogeneity can be detrimental to privacy protection techniques such as DP and secure aggregation because adding noises to update or aggregation processes may not be effective in protecting those clients who have highly skewed data distributions. Moreover, the issue can also have a detrimental effect on anomaly detection, making it more difficult to distinguish between malicious and benign, yet heterogeneous, updates.

System heterogeneity makes communication even less efficient by adding variability to client involvement and update latency. Clients with low bandwidth or processing capabilities may experience delays or dropouts during training, resulting in straggler effects in synchronous FL and outdated updates in asynchronous contexts. System heterogeneity can lead to an increase in communication rounds as a result of partial involvement and ineffective synchronization [181, 182]. To address these issues, people commonly use methods like asynchronous aggregation, client selection, and adaptive communication protocols. However, these methods come with trade-offs that affect fairness and staleness.

Model heterogeneity, referring to the utilization of varied model designs or capabilities by clients, also influences the effectiveness of communication. In these situations, classic parameter averaging doesn't work directly; hence, other methods like KD or partial model sharing are needed. HeteroFL [164] and recent advancements in heterogeneous FL systems demonstrate that transmitting only partial or compressed updates can diminish communication traffic; however, this complicates the maintenance of global model consistency.

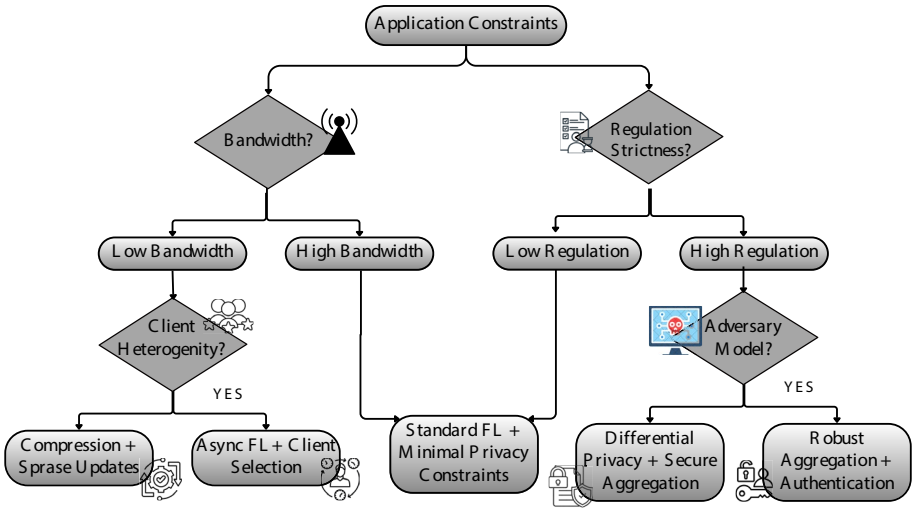


Fig. 9 Decision framework for selecting communication-efficient and privacy-preserving FL strategies based on system constraints and threat models

Overall, heterogeneity makes communication more challenging by slowing convergence, increasing variability in client updates, and complicating aggregation strategies. In addition to communication issues, heterogeneity also impacts privacy preservation, as Non-IID data distributions and diverse client behaviors can amplify privacy leakage risks and reduce the effectiveness of standard privacy mechanisms such as DP and secure aggregation. To address these challenges, a combination of approaches is required, including adaptive optimization, clustering, personalized FL, privacy-aware mechanisms, and hybrid communication strategies. Recent research trends focus on jointly managing statistical and system heterogeneity while balancing privacy, utility, and communication efficiency to develop scalable and robust FL frameworks suitable for real-world deployments.

8 Contextual analysis of communication threat defense trade-offs in FL

FL has various applications, including healthcare, finance, smart cities, edge -based IoT systems, etc. The privacy threats, defense mechanisms required to protect the sensitive data, bandwidth requirements, and model robustness in these scenarios vary according to the user needs. This section provides a context aware analysis and proposal for the application requirements, threat models, defense mechanisms, and their estimated communication and performance cost. The Fig. 9 illustrates how key application constraints in FL, such as bandwidth and regulation strictness, influence the selection of appropriate system strategies. Based on these conditions, solutions like compression, asynchronous training, DP, and robust aggregation are adopted to ensure efficient, secure, and privacy-preserving FL.

In healthcare systems such as a hospital, patient data storage is distributed in multiple locations. Pathology reports, image records, etc., are maintained in the respective laboratory units. The medical and genetic history is maintained in the administrative office. Disease diagnosis, personalized treatment plans, etc. are stored in the doctor's case sheet. In this scenario vertical FL will eliminate the need for combining records in a central repository. However, the threat of inference attack is high. DP or secure aggregation is the ideal defense mechanism, but they introduce distinct cost trade-offs.

DP mechanisms, particularly with strong privacy guarantees (low ϵ), inject noise into updates, leading to accuracy degradation and requiring additional communication rounds to converge. Secure aggregation, on the other hand, preserves model accuracy but increases communication overhead due to cryptographic operations and multi-round protocols, resulting in higher latency and payload expansion. Hierarchical FL or model compression techniques can be applied to enhance communication efficiency. however, compression may amplify reconstruction risks if not combined with privacy-preserving techniques. In the scenario where multiple hospitals are collaboratively training machine learning models, horizontal FL with model compression or local computation optimization is suitable. Local computation optimization reduces communication rounds, but excessive local updates under Non-IID data can lead to client drift, increasing overall communication cost. Reconstruction and integrity are possible threats in this scenario.

Modern smart homes equipped with IoT devices continuously derive privacy sensitive behavioral data. Cross-device FL can be applied for detecting anomalies, energy optimization, providing recommendations, etc. However, frequent communication of model updates exposes systems to reconstruction attacks. Hence, effective defense strategies such as DP, secure aggregation, or update sparsification must be carefully selected and tuned, balancing privacy guarantees with communication and computational efficiency.

In industrial environments which involve a large number of distributed sensors, robots and edge devices, decentralized FL is suitable to train machine learning models for process optimization and predictive maintenance. Decentralized FL avoids the problem of communication bottleneck which is essential in time critical systems. Event driven updates can be applied to reduce the communication latency without affecting the model robustness. Decentralized FL in industrial IoT systems is vulnerable to sybil and byzantine attacks. Robust aggregation techniques, defenses against sybil attacks and authentication protocols can be applied to secure the system. For instance, byzantine-resilient aggregation methods often require exchanging additional statistics or performing multiple validation steps, increasing communication cost. Furthermore, partial participation and asynchronous updates may amplify instability in adversarial environments.

Asynchronous FL with sparsification allows for low-latency communication and heterogeneous compute capacities, while SMPC protects feature privacy in financial fraud detection, where institutions store vertically partitioned data under stringent compliance. However, asynchronous updates introduce staleness, which can be exploited by adversaries. Similarly, HE and clustered updates protect against duplicate communication and collusion attacks, but significantly increase communication payload due to encrypted data exchange and multi-round interactions. Hierarchical FL aids in smart city traffic control by managing distributed nodes and reduces long-distance communication but introduces coordination overhead between layers. Customized FL is required for wearable health monitoring, as it supports user-specific data patterns, where sensitive vitals are protected by LDP and network transmission is minimized by local computation optimization. Last but not least, transfer FL facilitates knowledge exchange among companies with different data and model structures in collaborative drug discovery, while privacy amplification and layer-wise updating reduce transmission costs and mitigate the risks associated with property inference.

Recent advancements in FL have also been explored in the context of recommender systems, focusing on improving personalization, communication efficiency, and data privacy. For instance, a diffusion-based federated recommender system has been proposed to enhance recommendation quality through data augmentation and guided denoising mechanisms [183]. Similarly, reinforcement learning-based approaches such as FedRL [184] optimize communication efficiency by dynamically selecting participating clients and generating adaptive models. Personalized federated recommender systems have also been developed using fine-grained transformation and hybrid information sharing strategies to improve user-specific recommendations [185]. Furthermore, generative AI-based frameworks such as PIFGSR [186] integrate multi-source information using pluggable architectures to enhance recommendation performance. In addition, lightweight and efficient architectures, such as Slimify-based federated recommender systems [187], aim to reduce model complexity while maintaining accuracy.

9 Practical challenges in deploying communication-efficient and secure FL systems

Many methods have been suggested to make FL communication more efficient and protect privacy, but it is still hard to put them into practice in real-world systems. One big problem is that privacy-preserving technologies add extra work for computers and communication. For example, DP needs careful tuning of noise parameters since strong privacy assurances can make the model less accurate and require more communication cycles to reach convergence. Secure aggregation, HE, and SMPC necessitate additional communication and cryptography steps, which may be infeasible for edge devices with limited resources or latency-sensitive applications.

The heterogeneity of systems complicates deployment due to varying levels of computing power, network bandwidth, and availability among customers. Certain methodologies, including asynchronous training and client selection, may mitigate these challenges; however, they concurrently complicate issues related to stale updates, fairness, and instability in adverse conditions. Furthermore, communication-efficient techniques, such as sparsification and quantization, require careful calibration of their hyperparameters to achieve an appropriate trade-off between compression ratio and model performance. They may also exhibit behaviors that are difficult to anticipate when trained or deployed under Non-IID data distributions, which are prevalent in real-world settings. Another major challenge involves the integration and harmonization of heterogeneous methodologies within a single FL framework. Hybrid techniques that combine compression and privacy mechanisms appear convincing, but they don't always work well together. For instance, compression can make it harder to find anomalies in robust aggregates, and encryption systems can make it harder to do quick model upgrades or debugging. Real-world deployments must also follow rules set by the government. This is especially relevant in fields like healthcare and finance, where rigorous privacy rules may limit the types of algorithms and system architectures.

Lastly, scalability and system-level limits are significant issues. Large-scale FL deployments with thousands or millions of devices must deal with unstable connections, partial participation, and changing network conditions. Efficient coordination, fault tolerance, and adaptive communication mechanisms are necessary, but they are not used

enough in real life. These problems show that making useful FL systems needs not only new algorithms but also careful thought on system, regulatory, and operational limits.

10 Conclusion and future work

FL is regarded as a promising approach for distributed learning while maintaining privacy. This survey indicates that several issues remain unresolved due to the trade-offs among privacy, model performance, and communication efficiency. Some of the current solutions, such as DP, SMPC, and HE, as well as model compression techniques, frequently have opposing effects, such as increasing communication difficulty, decreasing accuracy, or complicating computations. These conflicts demonstrate the need for collaborative optimization frameworks that can simultaneously handle privacy, communication, and accuracy, even in situations where the system has practical limitations.

Adaptive and context-aware mechanisms that adjust the frequency of communication, the amount of compressed data, and the privacy budgets in accordance with the requirements of the system and the applications are limited. In the future, research should focus on communication techniques that are based on learning and adaptive privacy allocation, where resources are used in the best possible way based on the distribution of client data, network conditions, and threat models. Furthermore, the integration of individualized FL with communication-efficient methodologies is still inadequately investigated, especially in predominantly Non-IID environments.

Another interesting trend is the use of decentralized trust and coordination systems, like blockchain-based FL, which can make issues more open, easier to check, and stronger against adversaries. But adding blockchain to the mix adds further communication and latency overhead, which shows how important it is to have lightweight consensus methods and ledger designs that work well in FL contexts.

Recent progress in foundation models and large language model (LLM)-based agents also gives FL systems new possibilities. LLM agents can make intelligent coordination, dynamic client selection, and adaptive communication scheduling possible. However, they can also make communication scalability, privacy leakage, and model size limits more difficult. Lastly, there exists strong demand for system-level co-design that optimizes communication protocols, privacy measures, and model structures for large-scale use. It is still a problem to connect theoretical progress with real-world applications, especially in contexts with limited bandwidth and a wide range of devices.

Author contributions
Anuja Kunjumon and Greeshma Sarath conducted the literature review, analyzed and synthesized the information, and prepared the manuscript. Jyothisha J Nair supervised the work, provided guidance on structuring and content, and critically revised the manuscript. All authors (Anuja Kunjumon, Greeshma Sarath, and Jyothisha J Nair) read and approved the final version of the manuscript.

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